常见词汇

geometry 几何　　buckling (屈曲)　　specimen (示例, 样品)　　hinge on (取决于)

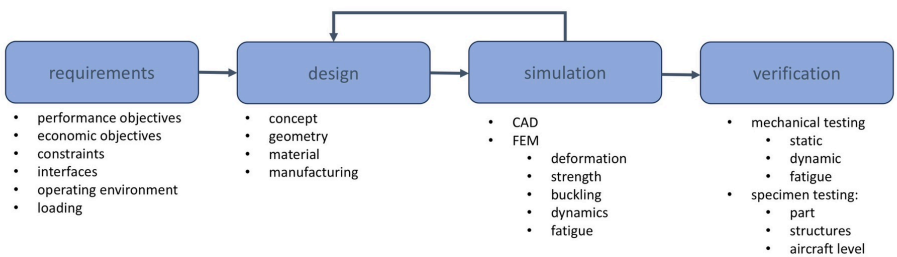
identity/diagonal/symmetric matrix　　clockwise/counter-clockwise 顺/逆时针
单位　对角　对称

magnituide 模 (matrix)　　vector/scalar 矢量/标量

1. introduction

基础回顾 repetition of fundamentals (statics, linear Algebra)
Concepts of elasticity (stress, equations of equilibrium, strain-displacement realtions, Hooke's law)
Material (propertities and behaviours of specific materials used in structure engineering)
Structure and Design Elements (link theoretial knowledge and real world structure)
Elasto-Stability (explore stability phenomena in light-weight structure)
Structure Analysis Methods (methods used to anahyse entire real-world structural problems)

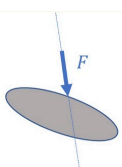
1. Lightweight Structural Design



In mechanics, effective problem-solving hinges on clear communication. Engineers must articulate the problem and its solution for both experts and non-experts. This clarity is also essential for one's **own understanding**. Though no single formula exists for every problem, a systematic approach is always required and typically follows these steps:

- Formulation of the engineering problem.
- Creating a simplified representation of a physical system / mechanical model and translate it into a mathematical problem.
- Solve the mathematical problem and transfer the results back to the mechanical domain.
- Discussion and interpretation of the results.

Classification of forces and cutting principle



force distribution (external force) ≠ stress (internal force)

Single Force: The idealization of a force acting at a specific point along a designated line of action is designated as a point force or single force.

Line Force: Forces that are continuously distributed along a line.

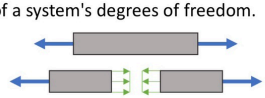
Surface Force: Forces that are continuously distributed on a surface, such as contact or hydrostatic pressure $\frac{F}{A}$.

Volume Force: Forces distributed over the volume of a body, such as gravitational forces $\frac{F}{V}$.

Applied Force: Loads such as forces or moments applied to a mechanical system.

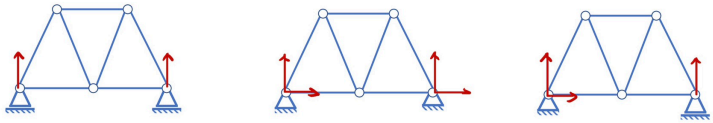
Reaction Force: Reaction forces (or constraint forces) are generated by the restriction of a system's degrees of freedom.

Inner Force: Internal forces between a system's components can be visualized by isolating a section and representing the forces on a free-body diagram (section cut).



A statically determinate system is one in which all support reactions and internal forces can be obtained exclusively from the equations of static equilibrium, without reference to material properties or deformation characteristics.

statically indeterminate systems | statically over-determinate systems | statically determinate systems



Framework 2D:

$2 \cdot (numbe\ of\ nodes) = (number\ of\ rods) + (number\ of\ reaction\ forces)$ < over-determinate

Framework 3D:

$3 \cdot (numbe\ of\ nodes) = (number\ of\ rods) + (number\ of\ reaction\ forces)$ > indeterminate

Linear algebra: A mathematical framework useful in engineering mechanics for understanding and solving a wide range of problems.

Vectors: Ordered lists of numbers: $x = [1 \quad 3 \quad 4]$

Matrices: Rectangular arrays of numbers, e.g., representing linear transformations, systems of equations, and data sets:

$$A = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 6 & 10 \\ 4 & 12 & 20 \end{bmatrix}$$

identity matrix: $$I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

diagonal matrix: $$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

symmetric matrix: $$S = \begin{bmatrix} 1 & 4 & 5 \\ 4 & 2 & 6 \\ 5 & 6 & 3 \end{bmatrix}$$

Transpose: Swapping between rows and columns:

$$x^T = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix} \qquad A^T = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 6 & 12 \\ 5 & 10 & 20 \end{bmatrix}$$

There are two forms of vector multiplication. The inner product, also known as the dot product or scalar product, is defined as follows

Scalar product: Takes two vectors and produces a single scalar number as its result, describing their relative direction and magnitudes.

$u \cdot v = [u_0 v_0 \quad u_1 v_1 \quad u_2 v_2 \quad \dots \quad u_n v_n]$

$u \cdot v = |u|\,|v| \cos(\theta)$

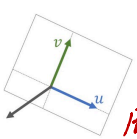
$\vec{u} \cdot \vec{v} = u_1 v_1 + u_2 v_2 + u_3 v_3$

Engineering Relevance:

- work done by a force,
- calculating angles between two vectors,
- vector projection

Cross or Vector product:

$$u \times v = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}$$



$\vec{u} \times \vec{v} = u_1 v_2 - u_2 v_1$

$|\vec{u} \times \vec{v}| = |\vec{u}||\vec{v}| \sin\theta$

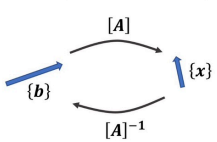
Engineering Relevance:

- moment and torque calculation
- normal vectors on the plane u and v

$$\vec{u} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2 v_3 - u_3 v_2)\vec{i} + (u_3 v_1 - u_1 v_3)\vec{j} + (u_1 v_2 - u_2 v_1)\vec{k}$$

Matrix Inversion: A matrix A represents a mapping from one vector to another. The inverse of a matrix A is the matrix that reverts vectors transformed by A back to their original vectors.

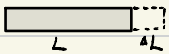
$[A]\{b\} = \{x\}$



For a matrix of size 2x2:

$$[A] = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \qquad [A]^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

2. Elasticity

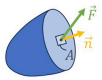


$\varepsilon = \frac{\Delta L}{L}$ ← deformation

↑ strain

1. Stress State

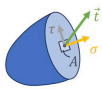
We begin by deriving the first set of equations, which describe the equilibrium between external and internal forces and stresses. Consider a deformable body, shown to the right, which is loaded with an external force **F** in its current deformed state. These loads can be applied as **surface forces** acting on the exterior of the body or as **body forces** distributed throughout its interior (e.g. gravity forces).



The traction vector at point ***P*** of a section with the area ***A*** is defined as:

Traction/stress vector: $$\vec{t} = \lim_{A \to 0} \frac{\boldsymbol{F}}{\boldsymbol{A}} = \frac{\boldsymbol{dF}}{\boldsymbol{dA}} \qquad (1)$$

The orientation of the **traction vector** depends on the cutting plane at point P, which means it depends on the orientation of the normal vector $\vec{\boldsymbol{n}}$.



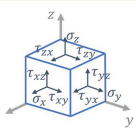
A linear mapping from one vector to another can be represented by a second-order tensor. This tensor is called the **Cauchy stress tensor** $\boldsymbol{\sigma}$:

$$\vec{t} = \boldsymbol{\sigma}\,\vec{n} \qquad (2)$$

The traction vector **t** can be decomposed into a normal stress σ, acting perpendicular to the plane, and a shear stress τ, acting parallel to the plane

$\vec{T} = \vec{\sigma_n} + \vec{\tau}$

1. Stress State



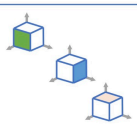
What information must $\boldsymbol{\sigma}$ contain to determine the traction vector?

Consider the three coordinate planes through a point:

x-plane $(\mathbf{n} = \mathbf{e}_x)$

y-plane $(\mathbf{n} = \mathbf{e}_y)$

z-plane $(\mathbf{n} = \mathbf{e}_z)$



$$\begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} \qquad (3)$$

$\vec{T} = \vec{\sigma} \cdot \vec{n}$

The stress tensor $\bar{\boldsymbol{\sigma}}$ relates the traction vector (force per unit area) acting on a surface to the orientation of the surface. If three perpendicular sections are selected in a x-y-z coordinate system, the second order tensor $\bar{\boldsymbol{\sigma}}$ can be expressed as:

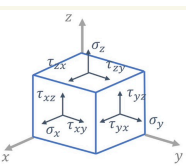
stress tensor: $$\bar{\boldsymbol{\sigma}} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$$

$\sigma(x,y,z)$ space position

dot → $\sigma(x,y,z)$

$\vec{n}$ ↓ surface (∞) → $\vec{t}$ → normal stress σ_n, shear stress τ

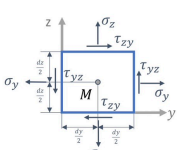
1. Stress State



By separating the stress tensor into its components, we have three normal stresses and six shear stresses. The normal stresses xx, yy , zz are often referred to in short as x, y, z. For the component indices, the first index describes the cutting plane, that is, in the direction of the surface normal, and the second index in the direction of the force component τ_{xy}.

force direction y
plane normal x

The 6 shear stresses are not all independent of each other. Calculating the moment equilibrium around the x-axis in the center *M*, it follows as an example for the shear stress τ_{zy}:



$\Sigma M: \; 2(-\tau_{zy} \cdot dx dy \cdot \frac{dz}{2}) + 2(\tau_{yz} \cdot dx dz \cdot \frac{dy}{2}) = 0$

$\tau_{zy} = \tau_{yz}$

moment equilibrium: $$2\frac{dy}{2}(\tau_{yz}\,dx\,dz) - 2\frac{dz}{2}(\tau_{zy}\,dx\,dy) = 0 \qquad (4)$$

Calculating the moment equilibrium around the corresponding other axes it can be seen

complementary shear stress : $$\tau_{xy} = \tau_{yx} \quad \tau_{xz} = \tau_{zx} \quad \tau_{yz} = \tau_{zy} \qquad (5)$$

Shear stresses in two sections perpendicular to each other are equal. These are referred to as **complementary shear stresses**.

Since the **complementary shear stresses** have the same sign, they are either pointing to the common edge of the cuboid or they are both directed away from it. Due to the stress relation, it follows that the components of the stress tensor can be reduced to **six independent** components.

stress tensor: $$\sigma = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{bmatrix} \quad (6)$$

$$\sigma_{ij} = \sigma_{ji} \quad (7)$$

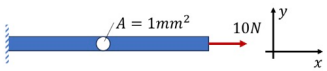
The general sign convention: By convention, a positive stress acts on a positive face in the positive coordinate direction. Consequently, a positive normal stress stretches the material (tension), while a negative normal stress compresses it

Summary:

1. Stress tensor -> has all the information $\vec{\sigma}$
2. Multiply by the plane normal -> traction vector on that plane. $\vec{t} = \vec{\sigma} \cdot \vec{n}$
3. Project onto the normal → normal stress $\vec{\sigma}_n = \vec{n}^T \cdot \vec{t}$
4. Subtract the projection → shear traction.
5. Take its magnitude → shear stress.

The Cauchy stress tensor contains all the information needed to determine the normal and shear stresses on any plane passing through a point.

Example: Rod under uniaxial tension.



$\sigma_x = \frac{N}{A} = 10\,MPa$

$\vec{\sigma} = \begin{bmatrix} -10 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ Cauchy Stress Tensor (3X3)

$\vec{t} = \vec{\sigma} \cdot \vec{n} = \begin{bmatrix} -10 \\ 0 \\ 0 \end{bmatrix}$ $\vec{n} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$ Traction (3X1)

$\sigma_N = \vec{n}^T \cdot \vec{t} = \vec{n}^T \cdot \vec{\sigma} \cdot \vec{n} = 10MPa$ Normal Stress (scalar)

$\vec{t}_n = \sigma_n \cdot \vec{n} = \begin{bmatrix} -10 \\ 0 \\ 0 \end{bmatrix}$ Normal Traction (3X1)

$\vec{t}_s = \vec{t} - \vec{t}_n = 0$ Shear Traction (3X1)

$\tau = |\vec{t}_s| = 0$ Shear Stress

Rotation

Rotation Matrix: $\bar{Q} = \begin{bmatrix} \cos45^\circ & \sin45^\circ & 0 \\ -\sin45^\circ & \cos45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Rotated Tensor: $\sigma' = \bar{Q} \sigma \bar{Q}^T$

$$\sigma' = \begin{bmatrix} \cos45^\circ & \sin45^\circ & 0 \\ -\sin45^\circ & \cos45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \cos45^\circ & -\sin45^\circ & 0 \\ \sin45^\circ & \cos45^\circ & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -5 & 0 \\ -5 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Internal Physical Loading: Blance normal force

$\det(\sigma - \lambda I) = 0$

$\vec{\sigma} = \begin{bmatrix} 10 & 0 \\ 0 & 0 \end{bmatrix}$

$\det(\vec{\sigma} - \lambda \vec{I}) = \begin{vmatrix} 10-\lambda & 0 \\ 0 & -\lambda \end{vmatrix} = -\lambda(10-\lambda) = 0$

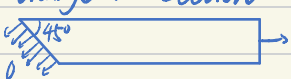
$\lambda_1 = 10 \quad \lambda_2 = 0$

$\vec{\sigma}' = \begin{bmatrix} 5 & -5 \\ -5 & 5 \end{bmatrix}$

$\det(\vec{\sigma}' - \lambda \vec{I}) = \begin{vmatrix} 5-\lambda & -5 \\ -5 & 5-\lambda \end{vmatrix} = (\lambda-5)^2 - 25 = 0$

$\lambda_1 = 10 \quad \lambda_2 = 0$

Change the section



$\vec{n} = [-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0]^T$

$\vec{t} = \vec{\sigma} \cdot \vec{n} = [-5\sqrt{2}, 0, 0]^T$

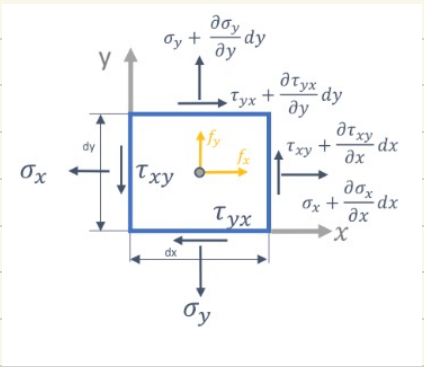
$\sigma_n = \vec{n}^T \cdot \vec{t} = 5MPa$ —— section A↑, stress distribution $\frac{F}{A}$↓

$\vec{t}_n = \sigma_n \cdot \vec{n} = [-\frac{5\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2}, 0]^T$

$\vec{t}_s = \vec{t} - \vec{t}_s = [-\frac{5\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2}, 0]^T$

$\tau = |\vec{t}_s| = 5MPa$

2. Equilibrium Equations



Taylor Expansion: $f(x+\Delta x) = f(x) + f'(x)\Delta x$

$\sigma_{x+dx} = \sigma_x + \frac{\partial \sigma_x}{\partial x} dx$

Force Equilibrium:

$$\begin{cases} \frac{\partial \sigma_x}{\partial x} + \frac{\partial \tau_{yx}}{\partial y} + \frac{\partial \tau_{zx}}{\partial z} + f_x = 0 \\ \frac{\partial \sigma_y}{\partial x} + \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{zy}}{\partial z} + f_y = 0 \\ \frac{\partial \sigma_z}{\partial z} + \frac{\partial \tau_{yz}}{\partial y} + \frac{\partial \tau_{xz}}{\partial x} + f_z = 0 \end{cases}$$

$$\text{div} -\vec{\nabla} \cdot \vec{\sigma} + \vec{f} = 0$$

$$\begin{bmatrix} \frac{d}{dx} & \frac{d}{dy} & \frac{d}{dz} \end{bmatrix} \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_z & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix} + \begin{bmatrix} f_x & f_y & f_z \end{bmatrix} = 0$$

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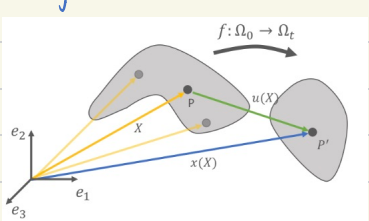
3. Deformation and Strain



Displacement : ΔL

Strain : $\varepsilon = \frac{\Delta L}{L}$

Reference Vector: $\bar{X} = X_1\bar{e}_1 + X_2\bar{e}_2 + X_3\bar{e}_3$

Deformed Configuration: $x(X)$

$\bar{u} = \bar{x} - \bar{X}$

Displacement function: $u(X)$

Rigid Body Translation: (14)

$$\bar{x} = \bar{X} + \bar{c} \qquad \bar{u} = \bar{x} - \bar{X} = \bar{c}$$

$\rightarrow \bar{c}$ is a constant vector

Rigid Body Rotation: (15)

$$\bar{x} = [Q]\bar{X} \qquad \bar{u} = \bar{x} - \bar{X} = (Q-1)\bar{X}$$

$$[Q] = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

Rigid Body Motions: (16)

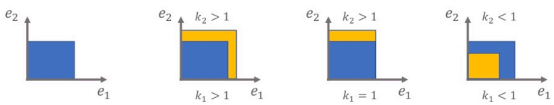
$$\bar{x} = [Q]\bar{X} + \bar{c}$$

$$\bar{u} = \bar{x} - \bar{X} = (Q-1)\bar{X} + \bar{c}$$

Note: In all 3 cases, the relative distance between the material points remains constant, therefore there are no strains!

Uniform Extension and Contraction: (27)

$$\bar{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{bmatrix} k_1 & 0 & 0 \\ 0 & k_2 & 0 \\ 0 & 0 & k_3 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \\ X_3 \end{Bmatrix} = \begin{Bmatrix} k_1 X_1 \\ k_2 X_2 \\ k_3 X_3 \end{Bmatrix} \qquad \bar{u} = \bar{x} - \bar{X} = \begin{Bmatrix} (k_1-1)X_1 \\ (k_2-1)X_2 \\ (k_3-1)X_3 \end{Bmatrix}$$

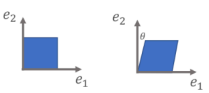


1. Contraction $0 < k < 1$
2. Expansion $k > 1$
3. No Change in length $k = 1$

Simple Shear: (17)

$$\bar{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{bmatrix} 1 & \tan\theta & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \\ X_3 \end{Bmatrix} = \begin{Bmatrix} X_1 + \tan\theta \cdot X_2 \\ X_2 \\ X_3 \end{Bmatrix}$$

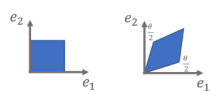
$$\bar{u} = \bar{x} - \bar{X} = \begin{Bmatrix} \tan\theta \cdot X_2 \\ 0 \\ 0 \end{Bmatrix}$$

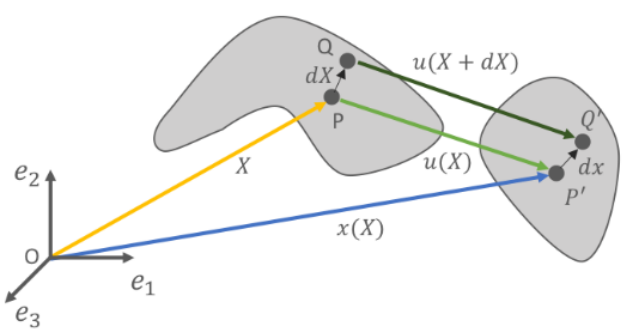


Pure Shear: (18)

$$\bar{x} = \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{bmatrix} 1 & \tan\theta/2 & 0 \\ \tan\theta/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{Bmatrix} X_1 \\ X_2 \\ X_3 \end{Bmatrix} = \begin{Bmatrix} X_1 + \tan\theta/2 \cdot X_2 \\ \frac{\theta}{2} \cdot X_1 + X_2 \\ X_3 \end{Bmatrix}$$

$$\bar{u} = \bar{x} - \bar{X} = \begin{Bmatrix} \tan\theta/2 \cdot X_2 \\ \tan\theta/2 \cdot X_1 \\ 0 \end{Bmatrix}$$





$P = X \qquad Q = X + dX$

$P' = X + u(X) \qquad Q' = (X + dX) + u(X + dX)$

$dx = \overline{P'Q'} = X + dX + u(X + dX) - X - u(X)$

$= dX + u(X + dX) - u(X)$

Taylor $\approx dX + u(X) + (\nabla u)dX - u(X)$

$$= (1 + \nabla u)dX$$

dx: deformation u: displacement

Gradient of the displacement function

$$\bar{H} = \nabla u = \frac{\partial u_i}{\partial X_j} = \begin{bmatrix} \frac{\partial u_1}{\partial X_1} & \frac{\partial u_1}{\partial X_2} & \frac{\partial u_1}{\partial X_3} \\ \frac{\partial u_2}{\partial X_1} & \frac{\partial u_2}{\partial X_2} & \frac{\partial u_2}{\partial X_3} \\ \frac{\partial u_3}{\partial X_1} & \frac{\partial u_3}{\partial X_2} & \frac{\partial u_3}{\partial X_3} \end{bmatrix}$$

$$\frac{\partial x_1}{\partial X_1} = \frac{\partial X_1 + \partial u_1}{\partial X_1} = 1 + \frac{\partial u_1}{\partial X_1} \qquad \frac{\partial x_1}{\partial X_2} = \frac{\partial X_1 + \partial u_1}{\partial X_2} = 0 + \frac{\partial u_1}{\partial X_2}$$

$$\bar{F} = \frac{dx_i}{dX_j} = \bar{I} + \nabla u = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} \frac{\partial u_1}{\partial X_1} & \frac{\partial u_1}{\partial X_2} & \frac{\partial u_1}{\partial X_3} \\ \frac{\partial u_2}{\partial X_1} & \frac{\partial u_2}{\partial X_2} & \frac{\partial u_2}{\partial X_3} \\ \frac{\partial u_3}{\partial X_1} & \frac{\partial u_3}{\partial X_2} & \frac{\partial u_3}{\partial X_3} \end{bmatrix} = \begin{bmatrix} \frac{\partial x_1}{\partial X_1} & \frac{\partial x_1}{\partial X_2} & \frac{\partial x_1}{\partial X_3} \\ \frac{\partial x_2}{\partial X_1} & \frac{\partial x_2}{\partial X_2} & \frac{\partial x_2}{\partial X_3} \\ \frac{\partial x_3}{\partial X_1} & \frac{\partial x_3}{\partial X_2} & \frac{\partial x_3}{\partial X_3} \end{bmatrix}$$

$$dx = (\bar{I} + \nabla u)\, dX = \bar{F}\, dX \qquad \boxed{\frac{dx}{dX} = \bar{F}}$$ Deformation Gradient

Length of dx: $dx = \bar{F}\, dX$

$$|dx|^2 = \bar{F}\, dX \cdot \bar{F}\, dX$$
$$= (\bar{F}\, dX)^T \cdot (\bar{F}\, dX) \quad (|A|^2 = A^T A)$$
$$= dX^T \cdot (\bar{F}^T \bar{F})\, dX$$

Length of dX: $|dX|^2 = dX^T \cdot \bar{I} \cdot dX$

Length comparision: $|dx|^2 - |dX|^2 = dX^T \cdot (\bar{F}^T \bar{F})\, dX - dX^T \cdot \bar{I} \cdot dX$
$$= dX^T (\bar{F}^T \bar{F} - \bar{I})\, dX$$

$$\boxed{\varepsilon_{Green} = \frac{1}{2}(\bar{F}^T \bar{F} - \bar{I}) = \frac{1}{2}(\bar{H} + \bar{H}^T + \bar{H}^T \bar{H})}$$

Green strain tensor (for finite deformation)

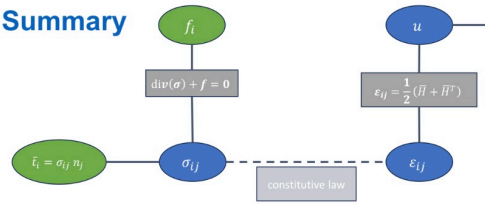
Small Strain Tensor: $\varepsilon_{Green} = \frac{1}{2}(\bar{H} + \bar{H}^T + \underbrace{\bar{H}^T \bar{H}}_{\ll \bar{H}}) = \boxed{\frac{1}{2}(\bar{H} + \bar{H}^T)}$

$$\varepsilon_{Green} = \begin{bmatrix} \varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_y & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_z \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) \\ \frac{1}{2}(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}) & \frac{\partial v}{\partial y} & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) \\ \frac{1}{2}(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}) & \frac{1}{2}(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}) & \frac{\partial w}{\partial z} \end{bmatrix}$$

1. Symmetric
2. Valid for small deformations

Engineering shear strain: $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

4. Summary



$$\varepsilon = \begin{bmatrix} \varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_y & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_z \end{bmatrix} = \frac{1}{2}(\bar{H} + \bar{H}^T) = \begin{bmatrix} \frac{\partial u}{\partial x} & \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) & \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \\ \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) & \frac{\partial v}{\partial y} & \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) \\ \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) & \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) & \frac{\partial w}{\partial z} \end{bmatrix}$$

$$\varepsilon_{Green} = \frac{1}{2}(F^T F - I) = \begin{bmatrix} \frac{\partial u}{\partial x} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial x}\right)^2\right] & \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) + \frac{1}{2}\left[\frac{\partial u}{\partial x}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\frac{\partial v}{\partial y} + \frac{\partial w}{\partial x}\frac{\partial w}{\partial y}\right] & \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) + \frac{1}{2}\left[\frac{\partial u}{\partial x}\frac{\partial u}{\partial z} + \frac{\partial v}{\partial x}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial x}\frac{\partial w}{\partial z}\right] \\ & \frac{\partial v}{\partial y} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right] & \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) + \frac{1}{2}\left[\frac{\partial u}{\partial y}\frac{\partial u}{\partial z} + \frac{\partial v}{\partial y}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\frac{\partial w}{\partial z}\right] \\ sym & & \frac{\partial w}{\partial z} + \frac{1}{2}\left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2\right] \end{bmatrix}$$

5.1 Law of Elasticity

homogeneous : properties are consistent throughout material

istropic : material's properties are independent of direction

$\sigma_{ij} \rightleftharpoons \varepsilon_{ij}$ $\quad \sigma_{ij} = C_{ijkl} \cdot \varepsilon_{kl}$ (elastic stiffness tensor) if linear, obey **Hooke's Law**

Voigt Notation

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{13} & \sigma_{32} & \sigma_{33} \end{bmatrix} \Rightarrow \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{bmatrix}$$

simplify calculation

Potential Energy : Π $\quad \sigma_{ij} = \frac{\partial \Pi}{\partial \varepsilon_{ij}}$ $\quad C_{klij} = \frac{\partial^2 \Pi}{\partial \varepsilon_{kl} \partial \varepsilon_{ij}}$, $C_{klij} = C_{ijkl}$ (Schwarz's theorem)

$$\begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{12} \\ \sigma_{13} \\ \sigma_{23} \end{bmatrix} = \begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & C_{1112} & C_{1113} & C_{1123} \\ & C_{2222} & C_{2233} & C_{2212} & C_{2213} & C_{2223} \\ & & C_{3333} & C_{3312} & C_{3313} & C_{3323} \\ & -sym- & & C_{1212} & C_{1213} & C_{1223} \\ & & & & C_{1313} & C_{1323} \\ & & & & & C_{2323} \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{13} \\ 2\varepsilon_{23} \end{bmatrix}$$

$\varepsilon_{ij} = \gamma_{ij}$

Stiffness Matrix 21 independent factor

5.2 Orthotropic Linear Elastic Material
(正交各向异性)

shear decoupling

$$\begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{13}\\ \sigma_{23}\end{Bmatrix} = \begin{bmatrix} C_{1111} & C_{1122} & C_{1133} & 0 & 0 & 0 \\ C_{1122} & C_{2222} & C_{2233} & 0 & 0 & 0 \\ C_{1133} & C_{2233} & C_{3333} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{1212} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{1313} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{2323} \end{bmatrix} \begin{Bmatrix}\varepsilon_{11}\\ \varepsilon_{22}\\ \varepsilon_{33}\\ 2\varepsilon_{12}\\ 2\varepsilon_{13}\\ 2\varepsilon_{23}\end{Bmatrix}$$

$$\begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{12}\\ 2\epsilon_{13}\\ 2\epsilon_{23}\end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{E_{11}} & -\frac{\nu_{21}}{E_{22}} & -\frac{\nu_{31}}{E_{33}} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_{11}} & \frac{1}{E_{22}} & -\frac{\nu_{32}}{E_{33}} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_{11}} & -\frac{\nu_{23}}{E_{22}} & \frac{1}{E_{33}} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{13}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{23}} \end{bmatrix}}_{\bar{S} \equiv \text{Compliance Matrix}} \begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{13}\\ \sigma_{23}\end{Bmatrix}$$

$\frac{V_{ij}}{E_{ii}} = \frac{V_{ji}}{E_{jj}}$

柔度矩阵 $S = C^{-1}$

$$\begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{31}\\ \sigma_{12}\end{Bmatrix} = \begin{bmatrix} \frac{1-\nu_{23}\nu_{32}}{E_{22}E_{33}\triangle} & \frac{\nu_{21}+\nu_{31}\nu_{23}}{E_{22}E_{33}\triangle} & \frac{\nu_{31}+\nu_{21}\nu_{32}}{E_{22}E_{33}\triangle} & 0 & 0 & 0 \\ \frac{\nu_{12}+\nu_{13}\nu_{32}}{E_{33}E_{11}\triangle} & \frac{1-\nu_{31}\nu_{13}}{E_{33}E_{11}\triangle} & \frac{\nu_{32}+\nu_{31}\nu_{12}}{E_{33}E_{11}\triangle} & 0 & 0 & 0 \\ \frac{\nu_{13}+\nu_{12}\nu_{23}}{E_{11}E_{22}\triangle} & \frac{\nu_{23}+\nu_{13}\nu_{21}}{E_{11}E_{22}\triangle} & \frac{1-\nu_{12}\nu_{21}}{E_{11}E_{22}\triangle} & 0 & 0 & 0 \\ 0 & 0 & 0 & G_{23} & 0 & 0 \\ 0 & 0 & 0 & 0 & G_{31} & 0 \\ 0 & 0 & 0 & 0 & 0 & G_{12} \end{bmatrix} \begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{31}\\ 2\epsilon_{12}\end{Bmatrix}$$

$$\triangle = \frac{1-\nu_{12}\nu_{21}-\nu_{23}\nu_{32}-\nu_{31}\nu_{13}-2\nu_{12}\nu_{23}\nu_{31}}{E_{11}E_{22}E_{33}}$$

2 | (anisotropic) ⟶ 9 (orthotropic)

Young's Modulus E_{ij} : $E = \frac{\sigma}{\varepsilon}$ → tension

Poisson's Ratio V_{ij} : $V_{ij} = -\frac{\varepsilon_{ii}}{\varepsilon_{jj}}$ (load i → deformation j)

Shear Modulus G_{ij}

$$\varepsilon_{yy} = \frac{1}{E}(\sigma_{yy} - V\sigma_{xx})$$

5.3 Isotropic Linear Elastic Material

$$\begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{12}\\ 2\epsilon_{13}\\ 2\epsilon_{23}\end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{E} & -\frac{\nu}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & \frac{1}{E} & -\frac{\nu}{E} & 0 & 0 & 0 \\ -\frac{\nu}{E} & -\frac{\nu}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G} \end{bmatrix}}_{\bar{S} \equiv \text{Compliance Matrix}} \begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{13}\\ \sigma_{23}\end{Bmatrix}$$

only E, V

$$\begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{12}\\ \sigma_{13}\\ \sigma_{23}\end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{E(1-\nu)}{(1-2\nu)(1+\nu)} & \frac{\nu E}{(1-2\nu)(1+\nu)} & \frac{\nu E}{(1-2\nu)(1+\nu)} & 0 & 0 & 0 \\ \frac{\nu E}{(1-2\nu)(1+\nu)} & \frac{E(1-\nu)}{(1-2\nu)(1+\nu)} & \frac{\nu E}{(1-2\nu)(1+\nu)} & 0 & 0 & 0 \\ \frac{\nu E}{(1-2\nu)(1+\nu)} & \frac{\nu E}{(1-2\nu)(1+\nu)} & \frac{E(1-\nu)}{(1-2\nu)(1+\nu)} & 0 & 0 & 0 \\ 0 & 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & 0 & G & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{bmatrix}}_{\bar{Q} \equiv \text{Stiffness Matrix}} \begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{12}\\ 2\epsilon_{13}\\ 2\epsilon_{23}\end{Bmatrix}$$

→ Material Constants reduced to Young's Modulus E and poisons ratio ν

→ $G = \frac{E}{2(1+\nu)}$

5.4 Plane Stress Condition

Plane Stress Condition: All the stress in the same plane $\Rightarrow \sigma_{33} = \sigma_{23} = \sigma_{13} = 0$

$$\sigma = \begin{bmatrix} \sigma_{11} & \sigma_{12} & 0 \\ \sigma_{12} & \sigma_{22} & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ \varepsilon_{33} \\ 2\varepsilon_{12} \\ 2\varepsilon_{13} \\ 2\varepsilon_{23} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & 0 & 0 \\ S_{21} & S_{22} & S_{23} & 0 & 0 & 0 \\ S_{31} & S_{32} & S_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & S_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & S_{66} \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ 0 \\ \sigma_{12} \\ 0 \\ 0 \end{bmatrix} \Rightarrow \varepsilon = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & 0 \\ \varepsilon_{12} & \varepsilon_{22} & 0 \\ 0 & 0 & \varepsilon_{33} \end{bmatrix}$$

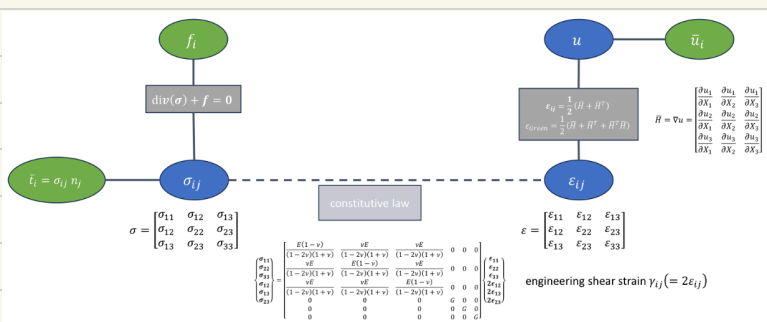
For orthotropic:

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_{12} \end{bmatrix} \qquad \varepsilon_{33} = -\left(\frac{\nu_{13}}{E_{11}}\sigma_{11} + \frac{\nu_{23}}{E_{22}}\sigma_{22}\right)$$

For isotropic:

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_{12} \end{bmatrix} \qquad \varepsilon_{33} = -\frac{\nu}{E}(\sigma_{11} + \sigma_{22})$$

$$Q = S^{-1} = \begin{bmatrix} \frac{E}{1-\nu^2} & \frac{E\nu}{1-\nu^2} & 0 \\ \frac{E\nu}{1-\nu^2} & \frac{E}{1-\nu^2} & 0 \\ 0 & 0 & \frac{1}{2(1+\nu)} \end{bmatrix}$$



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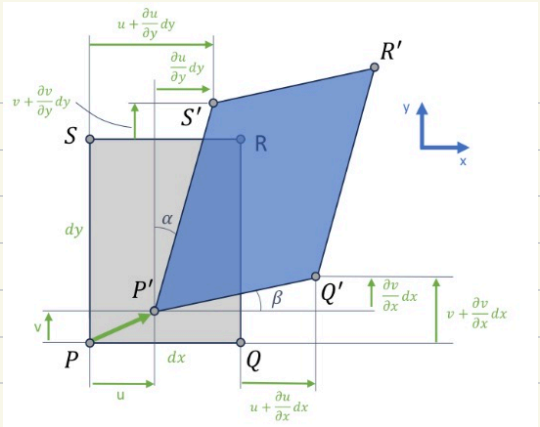
6. Energy Methods Lagrange's Equations

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{u}}\right) - \frac{\partial T}{\partial u} + \frac{\partial \Pi_i}{\partial u} = \frac{\partial W_e}{\partial u}$$

T: Kinematic energy $\quad \frac{1}{2}m\dot{u}^2$

Π_i: Potential energy $\quad \frac{1}{2}Ku^2$

Π_e: External work $\quad F \cdot u$



$$u(x+dx,y+dy)=u(x,y)+\frac{\partial u(x,y)}{\partial x}dx+\frac{\partial u(x,y)}{\partial y}dy$$
$$v(x+dx,y+dy)=v(x,y)+\frac{\partial v(x,y)}{\partial x}+\frac{\partial v(x,y)}{\partial y}dy$$

$$\overline{P'Q'}=dx+\frac{\partial u}{\partial x}dx$$
$$\overline{P'S'}=dy+\frac{\partial v}{\partial y}dy$$

$$\varepsilon_x=\frac{\overline{P'Q'}-\overline{PQ}}{\overline{PQ}}=\frac{\partial u}{\partial x}$$
$$\varepsilon_y=\frac{\overline{P'S'}-\overline{PS}}{\overline{PS}}=\frac{\partial v}{\partial y}$$

$$\tan\alpha=\frac{\frac{\partial u}{\partial y}dy}{dy+\frac{\partial v}{\partial y}dy}$$
$$\tan\beta=\frac{\frac{\partial v}{\partial x}dx}{dx+\frac{\partial u}{\partial x}dx}$$

small deformation: $\alpha=\tan\alpha=\frac{\frac{\partial u}{\partial y}dy}{dy+\frac{\partial v}{\partial y}dy}$ $\beta=\tan\beta=\frac{\frac{\partial v}{\partial x}dx}{dx+\frac{\partial u}{\partial x}dx}$

$\varepsilon_x,\varepsilon_y\ll 1$: $\alpha=\frac{\partial u}{\partial y}$ $\beta=\frac{\partial v}{\partial x}$

shear strain: $\gamma_{xy}=\alpha+\beta=\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}$

$$\varepsilon=\begin{bmatrix}\varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz}\\ \varepsilon_{yx} & \varepsilon_y & \varepsilon_{yz}\\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_z\end{bmatrix}=\begin{bmatrix}\varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz}\\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz}\\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z\end{bmatrix}$$

Strain: $\varepsilon_x=\frac{\partial u}{\partial x}$ $\varepsilon_y=\frac{\partial v}{\partial y}$ $\varepsilon_z=\frac{\partial w}{\partial z}$

shear strain: $\gamma_{xy}=\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}$ $\gamma_{xz}=\frac{\partial u}{\partial z}+\frac{\partial w}{\partial x}$ $\gamma_{yz}=\frac{\partial v}{\partial z}+\frac{\partial w}{\partial y}$

III. Structure Design Element

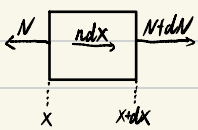
1: Load and Stress Types

- Tension and Compression
- Bending
- Shear
- Torsion

Normal Stress σ

Shear Stress τ

2. Truss Elements



$d \ll L$ (no sectional force)

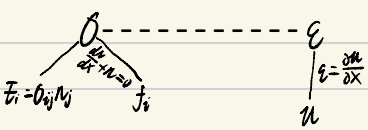
Tensile stiffness : $E \cdot A$

Equilibrium: $(N+dN)+ndx-N=0$

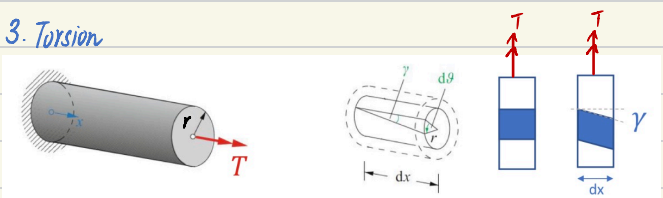
$$\frac{dN}{dx}+n=0$$

Kinematics : $\varepsilon=\frac{du}{dx}$

Constitutive : $\varepsilon=\frac{\sigma}{E}+\alpha_T \Delta T$



3. Torsion



Torsion Moment T

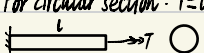
Torsional stiffness : GI_t

Shear Strain γ Twist Angle θ (rad) Shear Modulus G

Kinematic : $rd\theta=\gamma dx$, $\gamma=r\frac{d\theta}{dx}$

Constitutive law : $\tau=G\gamma=Gr\frac{\partial\theta}{\partial x}$

Equilibrium : $T=\int r\tau dA=G\frac{\partial\theta}{\partial x}\int r^2 dA$, For circular section : $T=GI_t\frac{d\theta}{dx}$



$d\theta=\frac{T}{GI_t}dx$

$\theta(l)-\theta(0)=\frac{Tl}{GI_t}$

$\theta_l=\frac{Tl}{GI_t}$ $\boxed{T=\frac{GI_t}{l}\theta_l}$

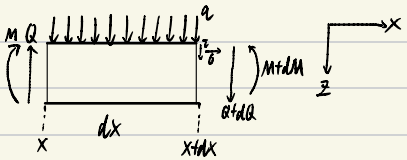
Thin-wall closed cross section

$I_t=\frac{4A_m^2}{\oint\frac{ds}{dt}}$ A_m: Section area, ds: unit length, t: wall thickness

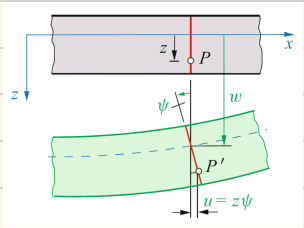
Thin-walled open cross section

$I_t=\frac{1}{3}\sum h_i t_i^3$ h_i: Length

4. Beam Elements



Equilibrium: $\frac{dQ}{dx} = -q \quad \frac{dM}{dx} = Q$



Kinematic: $\varepsilon = \frac{\partial u}{\partial x} \quad \gamma = \frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}$

Consitutive: $\sigma = E \cdot \varepsilon \quad \tau = G \cdot \gamma$

Bend Moment: $M = \int z \cdot \sigma(z) dA$

Shear Force: $Q = \int \tau(z) dA$

Normal Force: $N = \int \sigma(z) dA$

Beam theory assumptions: 1. $w = w(x)$ independent of z

$\varepsilon_z = \frac{\partial w}{\partial z} = 0$ · plane remains plane after deformation

2. $u(x,z) = z \cdot \psi(x) \Rightarrow \psi = \frac{\partial u}{\partial z}$

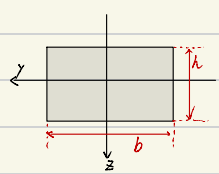
rotation angle $\psi(x)$

$\sigma = E \cdot \varepsilon = E \cdot \frac{\partial u}{\partial x} = E \cdot \frac{\partial \psi}{\partial x} \cdot z$

$M = \int z \cdot \sigma(z) dA = E\psi' \int z^2 dA$

$M = EI\psi' \quad \sigma = \frac{M}{I} z$

$\tau = G \cdot \gamma = G\left(\frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}\right) = G(w' + \psi)$



Moment of Area: $M_{pq} = \int_A y^p z^q dA$ (weight compared with area)

$p+q = 0/1/2$ (Order zero/one/two)

Cross section area (moment of area order zero)

$A = \int_A dA$

Calculation of CG (first order moment of Area)

$S_y = \int_A z dA \quad S_z = \int_A y dA \quad y_s = \frac{S_z}{A} \quad z_s = \frac{S_y}{A}$

Second moment of area

$I_y = \int_A z^2 dA \quad I_z = \int_A y^2 dA \quad I_{yz} = \int_A yz dA$

Rectangular cross section $I_y = \frac{bh^3}{12}$

Combined cross sections $I_y = \sum I_{y,i} + A_i \cdot z^2$

——Deflection Curve

Internal Force : $M=EI\psi'$ $Q=\kappa GA(w'+\psi)$ shear correction factor κ

Deformation : w, ψ $\kappa GA=GA_s$

Euler-Bernoulli Beam Assumption : 1. Plane remains plane 2. Always perpendicular to bending line

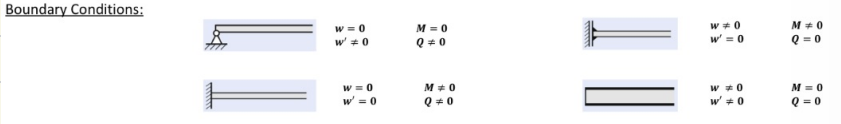
shear rigid (neglect the angular change of beam caused by shear force) :

$\kappa GA \to \infty \quad (w'+\psi)=\frac{Q}{\kappa GA}=0$

Beam cross sections keep perpendicular to bending line before and after deformation

$$w'+\psi=0 \Rightarrow w'=-\psi \xrightarrow{\psi'=\frac{M}{EI}} \boxed{w''=-\frac{M}{EI}} \xrightarrow{Q=M'} \boxed{Q=-(EIw'')'} \xrightarrow{Q'=-q} \boxed{q=(EIw'')''}$$

Boundary Conditions:



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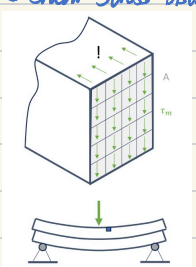
5. Shear Stress Distribution

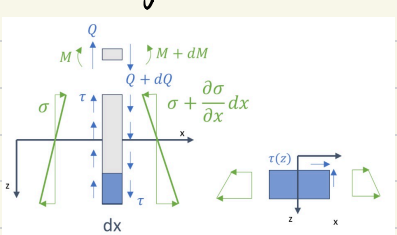


average shear force $\tau_m=\frac{Q_z}{A}$

Problem : $\tau_{xz}=\tau_m$, while $\tau_{zx}=0$ (applied on surface)

$\tau_{xz} \neq \tau_{zx}$

so the average distribution is incorrect



Equilibrium : $0=-\int\sigma dA+\tau(z)\cdot b(z)dx+\int(\sigma+\frac{\partial\sigma}{\partial x}dx)dA$

$\tau\cdot b=-\int\frac{\partial\sigma}{\partial x}dA$

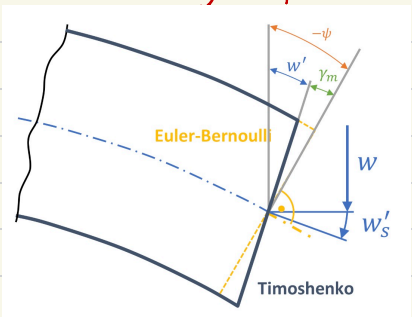
$0=\frac{M}{I_y}\cdot z \quad \frac{\partial\sigma}{\partial x}=\frac{\partial}{\partial x}(\frac{M}{I_y}\cdot z)=\frac{\partial M}{\partial x}\cdot\frac{z}{I_y}=\frac{Q}{I_y}\cdot z$

$\tau\cdot b=-\frac{Q}{I_y}\int z dA \quad S_y=\int z dA=\int_{z_1}^{z_m} z\cdot b\cdot dz$ (first moment of area)

$$\boxed{\tau(z)=\frac{Q_z}{I_y b}\cdot S_y}$$

6. Timoshenko Beam

Timoshenko Beam Theory Assumption: 1. Plane remains plane 2. $\psi + w' = \gamma_m$



Internal Forces: $M = EI\psi'$ $\quad Q = \kappa GA(w' + \psi) = GA_s(w' + \psi)$

Kinematics: $\psi + w' = \gamma_m$

$EI_y = \text{const}$ $\quad GA = \text{const}$

$\frac{dQ}{dx} = Q' = \kappa GA(\psi + w')' = -q$ $\quad \psi' + w'' = -\frac{q}{\kappa GA}$ $\quad \psi''' + w''' = -\frac{q''}{\kappa GA}$ $\quad \boxed{\kappa \approx \frac{5}{6}}$

Equilibrium: $M = EI_y\psi'$

$M' = Q = \kappa GA(\psi + w')$

$Q' = M'' = EI\psi''' = \kappa GA(\psi + w')'$

$EI_y\left(-\frac{q''}{\kappa GA} - w''''\right) = \kappa GA \cdot \left(-\frac{q}{\kappa GA}\right)$

$\boxed{EI_y w'''' = q - \frac{EI_y}{\kappa GA} q''}$ Deflection line after Timoshenko

bending deformation (Euler Beam)

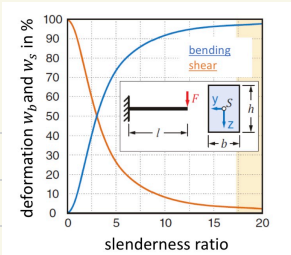
shear deformation (when $GA \to \infty$ is Euler Beam)

separate consideration of deformation components

$\frac{M}{EI_y} = \psi' = -w_b''$ bending deformation

$\frac{Q}{\kappa GA} = \psi + w' = w_s'(\gamma_m)$ shear deformation

$w_{total} = w_b + w_s$



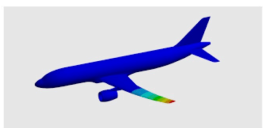
Timoshenko beam mainly consider:

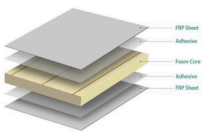
deformation change

stiffness χGA change → vibration W_0 resonance

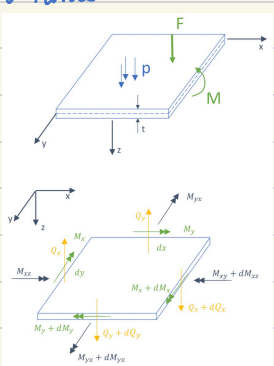
slenderness ratio: $\lambda = l \cdot \sqrt{\frac{A}{I_y}} = \frac{l}{h}\sqrt{12}$ $l = 5 \cdot h$

长细比





6. Plates



Transfer Loads: $Q_x = \int \tau_{xz} dz$ $Q_y = \int \tau_{yz} dz$

Bending moments: $M_x = \int \sigma_x \cdot z dz$ $M_y = \int \sigma_y \cdot z dz$

Torsion moments: $M_{xy} = M_{yx} \int \tau_{xy} \cdot z dz$

Equilibrium in z: $-p dx dy + (Q_y + \frac{\partial Q_y}{\partial y} dy) dx + (Q_x + \frac{\partial Q_x}{\partial x} dy) dx - Q_y dx - Q_x dy = 0$

$$\frac{\partial Q_x}{\partial x} + \frac{\partial Q_y}{\partial y} = -p$$

Equilibrium of moments: $\frac{\partial M_x}{\partial x} + \frac{\partial M_{yx}}{\partial y} = Q_x$ $\frac{\partial M_{xy}}{\partial x} + \frac{\partial M_y}{\partial y} = Q_y$

$\frac{\partial^2 M_x}{\partial x^2} + \frac{\partial^2 M_{yx}}{\partial x \partial y} = \frac{\partial Q}{\partial x}$ $\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = \frac{\partial Q_y}{\partial y}$

$$\frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} = -p$$

Displacement: $u(x,y,z) = z \cdot \psi_x(x,y)$ $v(x,y,z) = z \cdot \psi_y(x,y)$

Kinematic: $\varepsilon_x = \frac{\partial u}{\partial x} = z \cdot \frac{\partial \psi_x}{\partial x}$ $\gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = z\left(\frac{\partial \psi_x}{\partial y} + \frac{\partial \psi_y}{\partial x}\right)$

$\varepsilon_y = \frac{\partial v}{\partial y} = z \cdot \frac{\partial \psi_y}{\partial y}$ $\gamma_{xz} = \frac{\partial u}{\partial w} + \frac{\partial w}{\partial x} = \psi_x + \frac{\partial \psi_z}{\partial x}$

$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y} = \psi_y + \frac{\partial \psi_z}{\partial y}$

Constitutive law: $|\sigma_z| \ll |\sigma_x|, |\sigma_y| \Rightarrow |\sigma_z| \approx 0 \Rightarrow$ Hooke's Law in planes

$$\sigma_x = \frac{E}{1-\nu^2}(\varepsilon_x + \nu\varepsilon_y) = \frac{E}{1-\nu^2}\left(\frac{\partial \psi_x}{\partial x} + \nu\frac{\partial \psi_y}{\partial y}\right)z \qquad \tau_{xy} = G\cdot\gamma_{xy} = G\cdot\left(\frac{\partial \psi_x}{\partial y} + \frac{\partial \psi_y}{\partial x}\right)z$$

$$\tau_{xz} = G\cdot\gamma_{xz} = G\cdot\left(\psi_x + \frac{\partial w}{\partial y}\right)$$

$$\sigma_y = \frac{E}{1-\nu^2}(\varepsilon_y + \nu\varepsilon_x) = \frac{E}{1-\nu^2}\left(\frac{\partial \psi_y}{\partial y} + \nu\frac{\partial \psi_x}{\partial x}\right)z \qquad \tau_{yz} = G\cdot\gamma_{yz} = G\cdot\left(\psi_y + \frac{\partial w}{\partial x}\right)$$

Sectional force: $M_x = \int_{-\frac{t}{2}}^{\frac{t}{2}} \sigma_x \cdot z\,dz = \frac{E}{1-\nu^2}\left(\frac{\partial \psi_x}{\partial x} + \nu\frac{\partial \psi_y}{\partial y}\right)\int_{-\frac{t}{2}}^{\frac{t}{2}} z^2\,dz = \frac{Et^3}{12(1-\nu^2)}\left(\frac{\partial \psi_x}{\partial x} + \nu\frac{\partial \psi_y}{\partial y}\right)$

Plate stiffness $K = \frac{Et^3}{12(1-\nu^2)}$

$$M_x = K\left(\frac{\partial \psi_x}{\partial x} + \nu\frac{\partial \psi_y}{\partial y}\right) \qquad Q_x = G t_s\left(\psi_x + \frac{\partial w}{\partial x}\right)$$

$$M_y = K\left(\frac{\partial \psi_y}{\partial y} + \nu\frac{\partial \psi_x}{\partial x}\right) \qquad Q_y = G t_s\left(\psi_y + \frac{\partial w}{\partial y}\right)$$

$$M_{xy} = \frac{1-\nu}{2}K\left(\frac{\partial \psi_x}{\partial y} + \frac{\partial \psi_y}{\partial x}\right)$$

Reissner–Mindlin Plate Theory

Assume: $Gt_s \to \infty$

$$\psi_x + \frac{\partial w}{\partial x} = 0 \Rightarrow \psi_x = -\frac{\partial w}{\partial x} \qquad M_x = -K\left(\frac{\partial^2 w}{\partial x^2} + \nu\frac{\partial^2 w}{\partial y^2}\right)$$

$$\psi_y + \frac{\partial w}{\partial y} = 0 \Rightarrow \psi_y = -\frac{\partial w}{\partial y} \qquad M_y = -K\left(\frac{\partial^2 w}{\partial y^2} + \nu\frac{\partial^2 w}{\partial x^2}\right)$$

$$M_{xy} = -K(1-\nu)\frac{\partial^2 w}{\partial x \partial y}$$

$$K\left(\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4}\right) = p$$

Kirchhoff Plate Theory

Solution Methods:

1. Direct solution → special problems
2. Approximate solution → Fourier method or energy method
3. FEM

Fourier Method

Load: $q = q_0 \cdot \sin\left(\frac{\pi x}{l}\right)\cdot\sin\left(\frac{\pi y}{l}\right)$

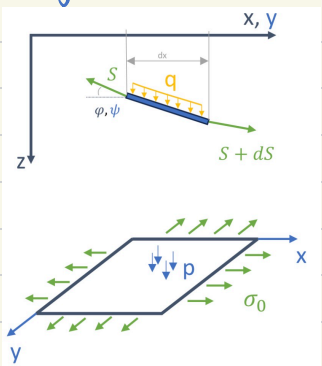
Approach: $w(x,y) = c\cdot\sin\left(\frac{\pi x}{l}\right)\cdot\sin\left(\frac{\pi y}{l}\right)$

BC's: $w(x=0,y) = w(x=l,y) = w(x,y=0) = w(x,y=l) = 0$

$$m_x(x=0,y) = m_x(x=l,y) = m_y(x,y=0) = m_y(x,y=l) = 0$$

Result: $c = \frac{q_0 l^4}{4\pi^4 K}$ $\quad w(x,y) = \frac{q_0 l^4}{4\pi^4 K}\cdot\sin\left(\frac{\pi x}{l}\right)\cdot\sin\left(\frac{\pi y}{l}\right)$

String and Membrane



Equilibrium in z: $(\sigma_0 t \frac{\partial \varphi}{\partial x} dx) dy + (\sigma_0 t \frac{\partial \psi}{\partial y} dy) dx + p dx dy = 0$

$\varphi = \frac{\partial w}{\partial x} \quad \psi = \frac{\partial w}{\partial y}$

$\sigma_0 t \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right) = -p$

Summary

1-Dimensional Design Elements

Truss: Hooke's law $\sigma = E\varepsilon$, $\sigma = \frac{N}{A}$, stiffness EA

Beam: Internal force $\sigma = \frac{M}{I} z^2$ → stress distribution above thickness, bending stiffness EI

Euler-Bernoulli Beam → rigid in shear → only bending deflection

Timoshenko Beam → finite shear → $w_{total} = w_b + w_s$

Torsion: Open/Closed cross-section

2-Dimensional Design Elements

Plates: Isotropic Plate bending stiffness $K = \frac{Et^3}{12(1-\nu^2)}$

Kirchhoff/Mindlin plate theory

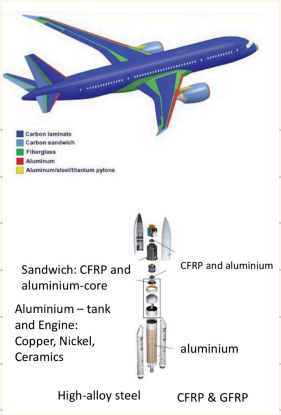
Fourier solution

7. Lightweight Structural Design

material properties | geometrical properties

tensile stiffness:	E	A
bending stiffness:	E	I_b
torsional stiffness:	G	I_t
shear stiffness:	G	A

IV. Lightweight Materials



1. Materials

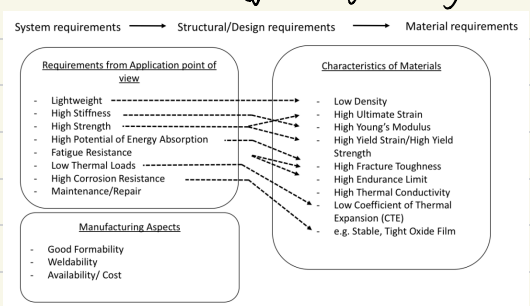
Typical Aerospace Materials

Mental Alloys: steel, aluminium, magnesium, titan

Non-Mental: composites, plastics, ceramics

eg. CFRP or GFRP

Material Nomenclature: stiffness, strength, density, costs, corrosion resistance, ductility, thermal expansion



environment objectives: lightweight structural design → fuel saving or more payload → less operating costs

performance objectives: for military or arcobatic aircrafts → maximum performance

operating time objectives: material and structural failure → strength → fatigue

⇓

Aircraft mission defines requirements for material selection

2.1 Steel Alloys

$\rho = 7.8\ kg/m^3$ $E \approx 210\ GPa$ Tensile Elastic Strength: $R_{p0.2} = 250 - 1200\ MPa$

stress causing 0.2% permanent plastic strain

high wear resistance	High density
high strength possible	corrision (expect for a few alloy)
hardenable (heat treatment)	
low price	
good fatigue behaviour	
well understood	

Landing Gear: Forged steel structures. High demand on strength (landing shock) and fatigue strength

Used for Components:

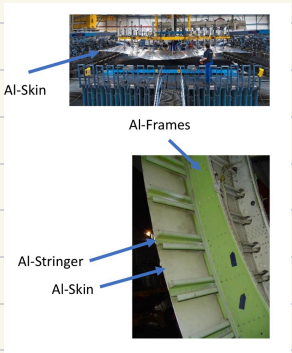
- lightweight structures that are subject to high mechanical loads
- high strength and stiffness requirements
- joiners with high bearing pressure (e.g. bolts)

Joiner (bolts, screws)

2.2 Aluminium Alloys $\rho = 2.7\ kg/m^3$ $E = 70GPa$ $R_{p0.2} = 50-500MPa$

most used material on aircraft (expect for newest material)

- low density
- simple manufacturing
- good resistance against stricks
- self protection against corrosion (oxide coating)
- electro conductive (lightning strick)
- good understanding of the material behaviour → impact of structural design
- low price
- recyclable

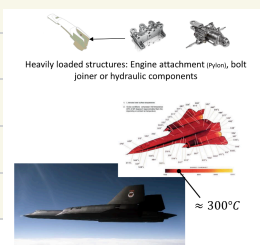


2.3 Titanium Alloys $\rho = 4.5 kg/m^3$ $E \approx 100-120GPa$ $R_{p0.2} = 200-1100MPa$

high strength	expensive
low decreasing strength at higher temperatures possible	bad processability
good corrosion resistance	Low Youngs Modules

Used for Components:

- with high temperature loading
- High structural loading with limited available space



Example SR-71: The strength of aluminium decreases with increasing temperatures - at a airspeed of $Ma \approx 2.0$ titan is used for aircraft design (friction heating)

2.4 Generalized Hooke's Law

Linear elastic behaviour, small deformations: → Hooke's Law: $\sigma_{ij} = C_{ijkl}\varepsilon_{ij} = \frac{Ev}{(1+v)(1-2v)}\delta_{ij}\delta_{kl} + \frac{E}{2(1+v)}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right)$

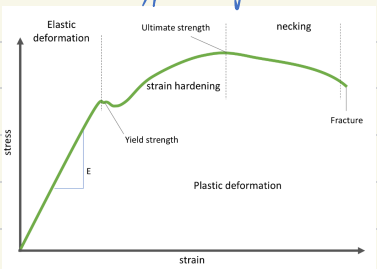
Matrix Notation:

$$\begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{31}\\ 2\epsilon_{12}\end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{1}{E} & -\frac{v}{E} & -\frac{v}{E} & 0 & 0 & 0 \\ -\frac{v}{E} & \frac{1}{E} & -\frac{v}{E} & 0 & 0 & 0 \\ -\frac{v}{E} & -\frac{v}{E} & \frac{1}{E} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G} \end{bmatrix}}_{[S] \equiv \text{Compliance Matrix}} \begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{31}\\ \sigma_{12}\end{Bmatrix} \quad \begin{Bmatrix}\sigma_{11}\\ \sigma_{22}\\ \sigma_{33}\\ \sigma_{23}\\ \sigma_{31}\\ \sigma_{12}\end{Bmatrix} = \underbrace{\begin{bmatrix} \frac{E(1-v)}{(1-2v)(1+v)} & \frac{vE}{(1-2v)(1+v)} & \frac{vE}{(1-2v)(1+v)} & 0 & 0 & 0 \\ \frac{vE}{(1-2v)(1+v)} & \frac{E(1-v)}{(1-2v)(1+v)} & \frac{vE}{(1-2v)(1+v)} & 0 & 0 & 0 \\ \frac{vE}{(1-2v)(1+v)} & \frac{vE}{(1-2v)(1+v)} & \frac{E(1-v)}{(1-2v)(1+v)} & 0 & 0 & 0 \\ 0 & 0 & 0 & G & 0 & 0 \\ 0 & 0 & 0 & 0 & G & 0 \\ 0 & 0 & 0 & 0 & 0 & G \end{bmatrix}}_{[Q] \equiv \text{Stiffness Matrix}} \begin{Bmatrix}\epsilon_{11}\\ \epsilon_{22}\\ \epsilon_{33}\\ 2\epsilon_{23}\\ 2\epsilon_{31}\\ 2\epsilon_{12}\end{Bmatrix}$$

Material Properties

	E	v	$G = \frac{E}{2(1+v)}$	ρ
Aluminium	70GPa	0.33	26.3GPa	$2.7 \times 10^{-9}\ t/mm^3$
Steel	210GPa	0.3	79.0GPa	$7.8 \times 10^{-9}\ t/mm^3$
Titanium	110GPa	0.31	42.0GPa	$4.5 \times 10^{-9}\ t/mm^3$

2.5 Failure Hypothesis for Mentals



Main types of failure: Brittle failure, Ductile failure, Fatigue failure

Maximum Normal Stress (Rankine): $\sigma_v = \max(\sigma_1, \sigma_2, \sigma_3)$ for brittle materials eg: ceramics, hardened steel

Maximum Shear Stress Theory (Tresca): $\sigma_v = \max(|\sigma_1-\sigma_2|, |\sigma_2-\sigma_3|, |\sigma_3-\sigma_1|)$ for ductile or brittle materials under multi-axial pressure

Maximum Disdistortion Energy Criterion (Von-Mises yield criterion): $\sigma_v = \sqrt{\frac{(\sigma_1-\sigma_2)^2+(\sigma_2-\sigma_3)^2+(\sigma_3-\sigma_1)^2}{2}}$ for ductile materials

σ_1: Maximum Priciple Stress σ_2: Intermediate Principle Stress σ_3: Minimum Priciple Stress $(\sigma_1 \geq \sigma_2 \geq \sigma_3)$

shear stress vanish for some orientations. Normal stress on these orientations are called Priciple Stress

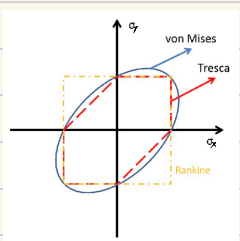
$\sigma = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}$ eigenvalue $\det(\sigma-\lambda I)=0 \Rightarrow \sigma_1, \sigma_2, \sigma_3$ $\sigma_{Priciple} = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$

For 2-Dimension plane $\sigma_{1,2} = \frac{\sigma_x+\sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}}$

Maximum Normal Stress (Rankine): $\sigma_v = \frac{\sigma_x+\sigma_y}{2} + \sqrt{\left(\frac{\sigma_x-\sigma_y}{2}\right)^2+\tau_{xy}}$

Maximum Shear Stress Theary (Tresca): $\sigma_v = \sqrt{(\sigma_x-\sigma_y)^2+4\tau_{xy}^2}$

Maximum Disdistortion Energy Criterion (Von-Mises yield criterion): $\sigma_v = \sqrt{\sigma_x^2-\sigma_x\sigma_y+\sigma_y^2+3\tau_{xy}^2}$

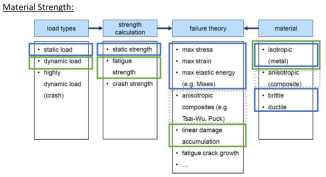


2.6 Fatigue Failure

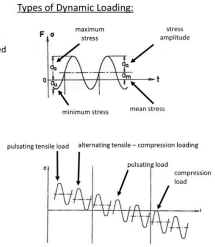
2. Dynamic Loading Classification

Static Loading: In reality there are no static loading. Loading is always time depended. Loading can be seen as quasi static if loading is acting very slow, so that time depended effect, e.g. inertia effect can be neglected.

Material Strength:

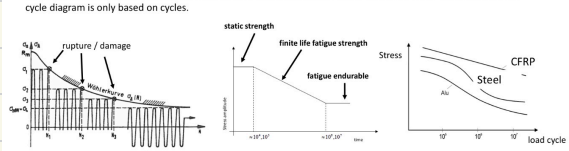


Types of Dynamic Loading:



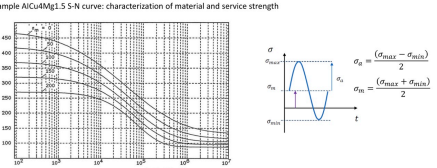
2.6 Woehler Diagram / Stress-Cycle Diagram

Testing of material and component behaviour under harmonic loading. Note: Frequency is not considered. Stress-cycle diagram is only based on cycles.



2.6 Woehler Diagram / Stress-Cycle Diagram

Example AlCu4Mg1.5 S-N curve: characterization of material and service strength

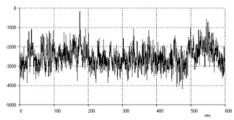


2.6 Dynamic Loading in Structural Applications

Load-Time Diagram: How to describe the time dependent loading in real applications:

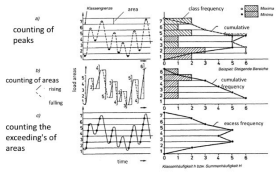
- Load over time can be measured?
- How often does a load level occur?

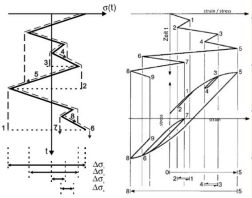
To analyse durability, we must characterize the load distribution.



2.6 Dynamic Loading in Structural Applications

Characterization of alternating loads by different counting methods.





Miner Rule / Cumulative Linear Damage Rule

A method to estimate fatigue life

Assume a linear accumulation of damage in the material

$$D = \sum_{i=1}^{K} \frac{n_i}{N_i} \leq 1.0$$

D: Fatigue Damage

K: the number of stress amplitude levels

n_i: the number of load cycles at level i

N_i: the maximum number of cycles at that particular amplitude (from S-N curve)

Summary

Metal Type	Stress State	Recommended Criterion
Ductile (steel, aluminium)	Uniaxial or Multiaxial	von-Mises (preferred), Tresca
Brittle (cast iron, concrete-like)	Uniaxial tension or bending	Maximum Normal Stress
Combined shear & tension	Multiaxial	von-Mise or Tresca
Fatigue-prone metals	Cyclic loading	S-N curve and accumulation rule

3. Composite Materials

composites from carbon-, glass- or aramid fiber embedded in a matrix material like epoxy, phenol or peek

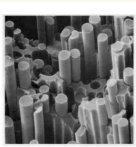
3.0 Composite Materials

Composite from carbon-, glass- or aramid fibre embedded in a matrix material like epoxy, phenol or peek.

+ low density with high strength and stiffness possible
+ high durability, low fatigue, crack-arresting properties of the laminate
+ anisotropic strength and stiffness of the fiber can be utilized by specifically orienting fibers within the component → many design degrees of freedom
+ Integral design possible
+ lots of possible combinations of fiber and matrix materials → many design degrees of freedom
- low impact resistance, damage not always visible from outside (e.g. delamination)
- complicated damage and failure mechanisms, degradation due to environmental influences (e.g. moisture absorption of the matrix)
- contact corrosion with other materials (e.g. carbon fiber with aluminum)
- no electrical conductivity (lightning strike → combination with copper mesh)
- limited operating temperature due to matrix material properties (mostly plastics, "*car in the sun*" → $T \approx 80°C$, → light paint colors)
- material is created during manufacturing of the component → high manufacturing effort, material properties depend on manufacturing, complex quality control

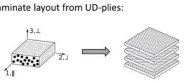
matrix material

fibre $\phi < 10\mu m$



Fracture surface, recorded with a scanning electron microscope

Laminate layout from UD-plies:



UD-plies with orthotropic material properties

Laminate layout with several UD-plies and different plie angle

T300: Typical CFRP UD-Prepreg ply for aerospace application

Ply thickness:	$t = 0.2mm$	$E_1 = 125GPa$	$G_{12} = 5GPa$
Density:	$\rho = 1550\ kg/m^3$	$E_2 = 8GPa$	$G_{31} = 5GPa$
Fibre volume fraction:	$V_f = 60\%$	$E_3 = 8GPa$	$G_{23} = 3.1GPa$

Fibers are the primary load-carrying component in composites, giving high stiffness and strength. Stiffness and strength vary with fiber type, diameter, and manufacturing process. Carbon fibers dominate high-performance structural applications due to high specific stiffness.

Fiber Type	Typical Young's Modulus (GPa)	Tensile Strength (MPa)	Comments
Glass fibers (E-glass, S-glass)	70 – 85 (E-glass) 85 – 100 (S-glass)	2000–3500	Low cost, corrosion resistant; widely used in marine, automotive
Carbon fibers (Standard, High Modulus)	230 – 300 (SM) 390 – 600 (HM)	3500 – 6000 (SM) 2500 – 4000 (HM)	High stiffness, low density; used in aerospace, sports equipment
Aramid fibers (Kevlar)	60–130	3000–3600	Very tough, impact resistant; low compressive strength, sensitive to UV
Natural fibers (flax, jute)	30–60	300–1000	Renewable, low cost, moderate strength; higher moisture absorption

E-glass (Electrical glass)

Alumino-borosilicate glass with lower magnesium and higher calcium content.

Originally designed for electrical insulation.

S-glass (Structural glass)

Magnesium aluminosilicate glass with lower calcium and higher silica and alumina content.

Designed for high strength structural applications.

Standard modulus (SM) carbon fiber:

SM fibres are more ductile, causing higher elongation before break. General applications are aerospace, automotive, wind turbine blades, and sports structures. Good balance of strength, stiffness, and cost.

High modulus (HM) carbon fiber:

HM fibers are stiffer but usually slightly weaker in tension compared to SM fibres. General applications are aerospace primary structures, space applications, precision instruments, satellites, and applications where stiffness-to-weight ratio is critical.

The matrix binds fibers together, transfers loads, and protects fibers from environmental damage. Common matrices are polymeric, metal, and ceramic. The matrix usually has much lower stiffness than the fiber, so the composite's properties are dominated by fiber properties, especially along the fiber direction.

Matrix Type	Typical Young's Modulus (GPa)	Tensile Strength (MPa)	Comments
Thermoset polymers (epoxy, polyester, vinyl ester)	2 – 4	40 – 100	Most common in composites; good chemical resistance, brittle
Thermoplastic polymers (PEEK, PPS, PP)	2 – 4	50 – 120	Tougher than thermosets; can be re-melted, higher processing temperatures
Metals (Aluminum, Titanium, Mg)	70 – 120	200 – 600	High toughness, higher density; used in MMCs (metal matrix composites)
Ceramics (SiC, Al2O3)	300 – 400	200 – 400	High stiffness, heat resistance; brittle, mainly in CMCs (ceramic matrix composites)

3.1 Mechanical Properties of UD-Piles

Volume Ratio :

$$\varphi_F = \frac{V_F}{V_C} = \frac{V_F}{V_F + V_m} \qquad \varphi_M = \frac{V_m}{V_C} = \frac{V_m}{V_F + V_m} \qquad \varphi_m + \varphi_F = 1$$

V: Volume F: Fibers M: Matrix C: Composites

Assumption:

Linear-elastic material behaviour

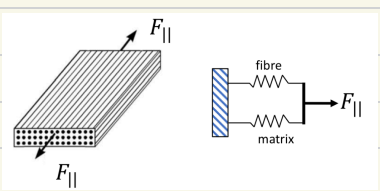
Macroscopically homogeneous

Homogeneous and isotropic matrix material

Homogeneous and isotropic fiber (glass) or orthotropic fiber (carbon, aramid fiber)

Perfect interface bonding

Fiber of uniform shape and size in a idealised arrangement



Voigt Model

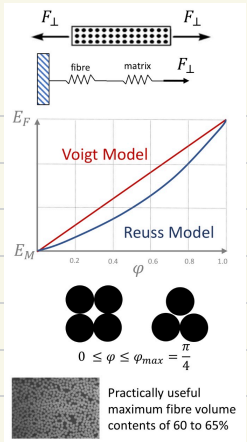
Force Equilibrium: $F_{11} = F_F + F_m \quad \sigma_{11} \cdot A_C = \sigma_F \cdot A_F + \sigma_M \cdot A_m$

Kinematic: $\varepsilon_{11} = \varepsilon_F = \varepsilon_m = \varepsilon_C$

Constitutive law: $\sigma_F = E_{F11} \cdot \varepsilon_F \quad \sigma_m = E_m \cdot \varepsilon_m$

$$E_{11} = \frac{E_F \cdot A_F + E_m \cdot A_m}{A_C} \xrightarrow[1-\varphi = \frac{A_m}{A_C}]{\varphi = \frac{A_F}{A_C}} E_{11} = E_F \cdot \varphi + E_m \cdot (1-\varphi)$$

Reuss Model (transverse stiffness)



Force equilibrium: $F_\perp = F_F = F_m \quad \sigma_c = \sigma_F = \sigma_m$

Kinematic: $\varepsilon_c = \varphi \cdot \varepsilon_F + (1-\varphi) \cdot \varepsilon_m$

$$\frac{\sigma_c}{E_\perp} = \varphi \cdot \frac{\sigma_F}{E_F} + (1-\varphi) \frac{\sigma_m}{E_m}$$

$$\frac{1}{E_\perp} = \varphi \cdot \frac{1}{E_F} + (1-\varphi) \frac{1}{E_m}$$

Halpin-Tsai (semi-empirical): $E_\perp = \frac{E_m (1 + \xi \cdot \eta \cdot \varphi)}{1 - \eta \cdot \varphi} \qquad \eta = \frac{(E_F/E_m - 1)}{E_F/E_m + \xi}, \quad \xi \approx 1$

Summary:

Young's Modulus in Fiber direction: $E_{11} = E_{F11} \cdot \varphi + E_m (1-\varphi)$

Transverse Young's Modulus: $\frac{1}{E_\perp} = \varphi \cdot \frac{1}{E_F} + (1-\varphi) \cdot \frac{1}{E_m}$

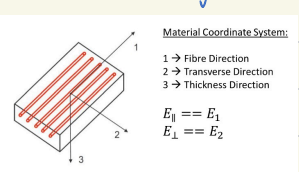
Shear Modulus: $G_{\perp 11} = G_m \frac{1}{(1-\varphi) + \frac{G_m}{G_{F,\perp 11}} \varphi}$

$$G_{\perp\perp} = \frac{E_\perp}{2(1+\nu_{\perp\perp})}$$

Possion's Ratio: $\nu_{11\perp} = \nu_F \cdot \varphi + \nu_m (1-\varphi)$

$$\nu_{\perp 11} = \nu_{11\perp} \frac{E_\perp}{E_{11}} \quad \left(\nu_{12} = \nu_{21} \cdot \frac{E_1}{E_2}\right)$$

3.2 Constitutive Law of UD-Piles



Material Coordinate System:

1 → Fibre Direction
2 → Transverse Direction
3 → Thickness Direction

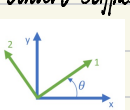
$E_\parallel == E_1$
$E_\perp == E_2$

Plane stresses $\sigma_{33} = \sigma_{23} = \sigma_{13} = 0$

$$\begin{bmatrix} \varepsilon_{11} \\ \varepsilon_{22} \\ 2\varepsilon_{12} \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{21} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix} \cdot \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{bmatrix} \qquad \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \cdot \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} \qquad \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix} = \begin{bmatrix} \frac{E_1}{1-\nu_{12}\nu_{21}} & \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{21}E_1}{1-\nu_{12}\nu_{21}} & \frac{E_2}{1-\nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{bmatrix} \cdot \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ 2\varepsilon_{12} \end{bmatrix}$$

Compliance Matrix [S]

Convert stiffness matrix from local (1-2) to globable (x-Y) coordinate system



$$R(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \qquad R^{-1} = R^T$$

Rotating the stress tensor:

$$\vec{t}_x = \vec{\sigma}_{xy} \cdot \vec{n}_x$$

vector rotation:

$$\vec{t}_{12} = \vec{R} \cdot \vec{t}_x \qquad \vec{t}_x = \vec{R}^T \cdot \vec{t}_{12}$$

$$\vec{n}_{12} = \vec{R} \cdot \vec{n}_x \qquad \vec{n}_x = \vec{R}^T \cdot \vec{n}_{12}$$

$$\vec{R}^T \cdot \vec{t}_{12} = \vec{\sigma}_{xy} \cdot \vec{R}^T \cdot \vec{n}_{12}$$

$$\vec{t}_{12} = \vec{R} \cdot \vec{\sigma}_{xy} \cdot \vec{R}^T \cdot \vec{n}_{12}$$

$$\vec{\sigma}_{12} = \vec{R} \cdot \vec{\sigma}_{xy} \cdot \vec{R}^T$$

$$\vec{\sigma}_{xy} = \vec{R}^T \cdot \vec{\sigma}_{12} \cdot \vec{R}$$

$$\sigma_x = \sigma_1 \cos^2\theta + \sigma_2 \sin^2\theta - 2\tau_{12} \sin\theta\cos\theta$$

$$\sigma_y = \sigma_1 \sin^2\theta + \sigma_2 \cos^2\theta + 2\tau_{12} \sin\theta\cos\theta$$

$$\tau_{xy} = \sigma_1 \sin\theta\cos\theta - \sigma_2 \sin\theta\cos\theta + \tau_{12}(\cos^2\theta - \sin^2\theta)$$

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = [T_\sigma]^{-1} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{bmatrix}$$

$$[T_\sigma]^{-1} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & 2\sin\theta\cos\theta \\ \sin\theta\cos\theta & -\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$$[T_\sigma] = \begin{bmatrix} \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$$\begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix} = [T_\varepsilon]^{-1} \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \gamma_{12} \end{bmatrix}$$

$$[T_\varepsilon]^{-1} = \begin{bmatrix} \cos^2\theta & \sin^2\theta & -\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & \sin\theta\cos\theta \\ 2\sin\theta\cos\theta & -2\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

$$[T_\varepsilon] = \begin{bmatrix} \cos^2\theta & \sin^2\theta & \sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -\sin\theta\cos\theta \\ -2\sin\theta\cos\theta & 2\sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{bmatrix}$$

How is a stiffness matrix converted from a local (1-2) to a global (x-y) coordinate system?

Step 1: (1)

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = [Q] \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{Bmatrix}$$

Step 2: Transform the stress from local to global coordinates. (2)

$$\{\sigma\}_1 = [T_\sigma]\{\sigma\}_x \qquad \{\sigma\}_x = [T_\sigma]^{-1}\{\sigma\}_1$$

Step 3: (1) → (2) (3)

$$\{\sigma\}_x = [T_\sigma]^{-1}[Q]\{\varepsilon\}_1$$

Step 4: Transform the strain from local to global coordinates. (4)

$$\{\varepsilon\}_1 = [T_\varepsilon]\{\varepsilon\}_x \qquad \{\varepsilon\}_x = [T_\varepsilon]^{-1}\{\varepsilon\}_1 \qquad \{\varepsilon\}_1 = \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{Bmatrix} \quad \{\varepsilon\}_x = \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

Step 5: (4) → (3) (5)

$$\{\sigma\}_x = [T_\sigma]^{-1}[Q][T_\varepsilon]\{\varepsilon\}_x$$

$$[\bar{Q}] = [T_\sigma]^{-1}[Q][T_\varepsilon]$$

alternative approach: $[T] = [T_\sigma]$

$$\begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} = [T] \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} \qquad \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_{12} \end{Bmatrix} = [T] \begin{Bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_{xy} \end{Bmatrix}$$

$$\begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{Bmatrix} = \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ 2\epsilon_{12} \end{Bmatrix} = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_{12} \end{Bmatrix}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = [T]^{-1}[Q] \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix} = [T]^{-1}[Q] \begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix} \begin{Bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_{12} \end{Bmatrix}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = [T]^{-1}[Q] \begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix} [T] \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_{xy} \end{Bmatrix}$$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = [T]^{-1}[Q] \underbrace{\begin{bmatrix} 1 & & \\ & 1 & \\ & & 2 \end{bmatrix} [T] \begin{bmatrix} 1 & & \\ & 1 & \\ & & \frac{1}{2} \end{bmatrix}}_{[T_\varepsilon]} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{Bmatrix}$$

Rotted stiffness matrix $[\bar{Q}]$ of the plane UD single layer in the global coordinate system:

$$[\bar{Q}] = [T][Q][T]^T$$

with:
$$[T] = \begin{bmatrix} cos^2(\theta) & sin^2(\theta) & -2\sin(\theta)\cos(\theta) \\ sin^2(\theta) & cos^2(\theta) & 2\sin(\theta)\cos(\theta) \\ \sin(\theta)\cos(\theta) & -\sin(\theta)\cos(\theta) & cos^2(\theta) - sin^2(\theta) \end{bmatrix}$$

3.3 Classical Laminate Theory

Assumptions:

Perfect Bonding

Homogeneous Layers

Plane Stress

Kirchhoff Hypothesis: stays perpendicular

Small Displacements

(always linear)

Kirchhoff Plate Theory: $\begin{bmatrix}\varepsilon_x\\ \varepsilon_y\\ \gamma_{xy}\end{bmatrix} = \begin{bmatrix}\varepsilon_x^0\\ \varepsilon_y^0\\ \gamma_{xy}^0\end{bmatrix} + z\begin{bmatrix}\kappa_x\\ \kappa_y\\ \kappa_{xy}\end{bmatrix}$ $\kappa_x = -\frac{\partial^2 w}{\partial x^2}$ $\kappa_y = -\frac{\partial^2 w}{\partial y^2}$ $\kappa_{xy} = -\frac{\partial^2 w}{\partial x \partial y}$

$\sigma_{xy} = \bar{Q}\cdot\varepsilon_{xy}$

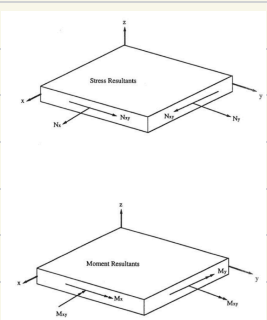
$\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xy}\end{bmatrix} = \begin{bmatrix}\bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{13}\\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{16}\\ \bar{Q}_{16} & \bar{Q}_{16} & \bar{Q}_{66}\end{bmatrix}\begin{bmatrix}\varepsilon_x^0\\ \varepsilon_y^0\\ \gamma_{xy}^0\end{bmatrix} + z\begin{bmatrix}\bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16}\\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26}\\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66}\end{bmatrix}\begin{bmatrix}\kappa_x\\ \kappa_y\\ \kappa_{xy}\end{bmatrix}$

$\varepsilon_x^0/\varepsilon_y^0$: tension on X/Y axis } in-plane

γ_{xy}^0: in-plane shear deformation } in-plane

κ_x/κ_y: bending on X/Y axis } bending

κ_{xy}: twist



Force Equilibrium: $N_x = \int_{-\frac{h}{2}}^{\frac{h}{2}} \sigma_x dz$ $N_y = \int_{-h/2}^{h/2} \sigma_y dz$ $N_{xy} = \int_{-h/2}^{h/2} \tau_{xy} dz$

Moment Equilibrium: $M_x = \int_{-h/2}^{h/2} \sigma_x z dz$ $M_y = \int_{-h/2}^{h/2} \sigma_y z dz$ $M_{xy} = \int_{-h/2}^{h/2} \tau_{xy} z dz$

$\begin{bmatrix}N_x\\ N_y\\ N_{xy}\end{bmatrix} = \int_{-h/2}^{h/2}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xz}\end{bmatrix} dz$ $\begin{bmatrix}M_x\\ M_y\\ M_{xy}\end{bmatrix} = \int_{-h/2}^{h/2}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \sigma_x z\end{bmatrix} z dz$

Layers are not continus, so integrate each layer and sum up

$\begin{bmatrix}N_x\\ N_y\\ N_{xy}\end{bmatrix} = \sum_{k=1}^{n}\int_{h_{k-1}}^{h_k}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xz}\end{bmatrix} dz$ $\begin{bmatrix}M_x\\ M_y\\ M_{xy}\end{bmatrix} = \sum_{k=1}^{n}\int_{h_{k-1}}^{h_k}\begin{bmatrix}\sigma_x\\ \sigma_y\\ \tau_{xz}\end{bmatrix} z dz$

(10) → (12):

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \int_{h_{k-1}}^{h_k} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix} z \, dz$$

$$\begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \int_{h_{k-1}}^{h_k} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} z + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix} z^2 \, dz$$

Bringing the constants to the front of the integral:

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix}$$

$$\begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix}$$

Performing the integration:

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} (h_k - h_{k-1}) + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix} \frac{1}{2}\left(h_k{}^2 - h_{k-1}{}^2\right)$$

$$\begin{Bmatrix} M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \sum_{k=1}^{n} \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \end{Bmatrix} \frac{1}{2}\left(h_k{}^2 - h_{k-1}{}^2\right) + \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{12} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \begin{Bmatrix} \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix} \frac{1}{3}\left(h_k{}^3 - h_{k-1}{}^3\right)$$

Laminate Stiffness Matrix – ABD Matrix:

$$A_{ij} = \sum_{k=1}^{n} [\bar{Q}_{ij}]_k (h_k - h_{k-1})$$

$$B_{ij} = \frac{1}{2} \sum_{k=1}^{n} [\bar{Q}_{ij}]_k \left(h_k^2 - h_{k-1}^2\right)$$

$$D_{ij} = \frac{1}{3} \sum_{k=1}^{n} [\bar{Q}_{ij}]_k \left(h_k^3 - h_{k-1}^3\right)$$

$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{Bmatrix} = \left[\begin{array}{ccc:ccc} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ \hdashline B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{array}\right] \begin{Bmatrix} \varepsilon_x^0 \\ \varepsilon_y^0 \\ \gamma_{xy}^0 \\ \kappa_x \\ \kappa_y \\ \kappa_{yx} \end{Bmatrix}$$

$$\begin{Bmatrix} N \\ M \end{Bmatrix} = \begin{bmatrix} A & B \\ B & D \end{bmatrix} \begin{Bmatrix} \varepsilon^0 \\ \kappa \end{Bmatrix}$$

in-plane stiffness; in-plane / bending coupling; bending stiffness

Stacking Sequence Notation:

Laminate nomenclature is a standardized system for identifying the fiber orientation of individual layers within a composite material. This nomenclature provides a concise representation of the laminate's overall configuration. The following outlines the key steps involved in designating a laminate

1. The layer sequence begins from the top surface of the laminate.
2. The stacking sequence specifies the fibre orientation relative to the global coordinate system, measured in degrees.
3. The stacking sequence is enclosed within square brackets **[ ]**.
4. Distinct layers or groups of layers are separated by a slash **/** or **,** symbol.
5. For repeated groups or layers, a subscript **n** is used to indicate the number of repetitions.
6. A symmetric laminate is indicated by a subscript **S** placed outside the square brackets $[\,]_S$

Examples:

$[45 \,/\, 30 \,/\, 0]$	$[45_2]$	$[(30)_2 \,/\, 0^\circ]$	$[45 \,/\, 30]_S$	$[\pm 45]_S$
3 plies of $45^\circ, 30^\circ, 0^\circ$	2 plies of 45°	3 plies of $30^\circ, 30^\circ, 0^\circ$	4 plies of $45^\circ, 30^\circ, 30^\circ, 45^\circ$	2 plies of $+45^\circ \mid -45^\circ$

Symmetric : $B = 0$

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} A & 0 \\ 0 & D \end{bmatrix} \begin{bmatrix} \varepsilon^0 \\ \kappa \end{bmatrix}$$

$$N = A\varepsilon^0$$

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{bmatrix} = \frac{1}{t} A \begin{bmatrix} \varepsilon_x \\ \varepsilon_y \\ \gamma_{xy} \end{bmatrix}$$

Engineering Constants:

Engineering constants can be calculated for two distinct loading scenarios: membrane and bending loading. For the membran condition, only the A-matrix is used, which is allowed for symmetrical laminates. The current objective is to derive an equation for stress that aligns with the uniaxial elasticity law. To achieve this, the equation is normalized by the total laminate thickness t. This process effectively homogenizes the material, resulting in the following expression.

Case 1: limiting the transverse contraction → $\varepsilon_y = \gamma_{xy} = 0$

$$\begin{Bmatrix} \sigma_x \\ \sigma_y \\ \tau_{xy} \end{Bmatrix} = \frac{1}{t_{total}} \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix} \begin{Bmatrix} \varepsilon_x \\ \varepsilon_y = 0 \\ \gamma_{xy} = 0 \end{Bmatrix} \qquad \sigma_x = \frac{A_{11}}{t_{total}} \varepsilon_x$$

$$E_x = \frac{A_{11}}{t_{total}} \qquad G_{xy} = \frac{A_{66}}{t_{total}}$$

$$E_y = \frac{A_{22}}{t_{total}}$$

Case 2: no limiting of transverse contraction → $\sigma_y = \tau_{xy} = 0$

$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_{xy} \end{Bmatrix} = t_{total} \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^{-1} \begin{Bmatrix} \sigma_x \\ \sigma_y = 0 \\ \tau_{xy} = 0 \end{Bmatrix} \qquad \sigma_x = \frac{1}{(A^{-1})_{11} t_{total}} \varepsilon_x$$

$$E_x = \frac{1}{(A^{-1})_{11} \cdot t_{total}} \qquad G_{xy} = \frac{1}{(A^{-1})_{66} \cdot t_{total}}$$

$$E_y = \frac{1}{(A^{-1})_{22} \cdot t_{total}}$$

Based on the stacking sequence, composite laminates can be categorized as:

Angle-ply Laminate
A laminate is called an angle-ply laminate if it has layers of the same thickness and material, and are oriented at θ and −θ.

Cross-ply Laminate
A laminate is called a cross-ply laminate if all the layers have the orientation 0° and 90°.

Balanced Laminate
A laminate is called a balanced laminate when it has pairs of layers of the same thickness and material, and the angles of the layers have opposite signs. Balanced laminates can also have layers at 0° and 90°.

Symmetric Laminate
A laminate is called symmetric when the material, angle, and thickness of the layers are the same above and below the midplane.

Antisymmetric Laminate
A laminate is called antisymmetric when the material and thickness of the layers are the same above and below the midplane, but the orientation of the layers have opposite signs above and below the midplane.

A_{16}, A_{26} : shear-tension coupling D_{16}, D_{26} : bending-twist coupling

Angle-ply Laminate

$[+\theta/-\theta]$

$A_{16}=A_{26}=0$: the shear-tension coupling caused by opposite angle cancelled out each other

Cross-ply Laminate

only 0°, 90° $\Rightarrow \bar{Q}_{16}=\bar{Q}_{26}=0$

$A_{16}=A_{26}=0$, $B_{16}=B_{26}=0$, $D_{16}=D_{26}=0$

no shear-tension and bending-twist coupling

Balance Laminate

For every θ° exists a $-\theta^\circ$ ($45^\circ/-45^\circ$, $0^\circ/90^\circ$), but don't need to neighbor or symmetric

$A_{16}=A_{26}=0$, no shear-tension coupling

Symmetric Laminate

$[\theta_1/\theta_2]_s$

$B=0$, no in-plane/bending coupling

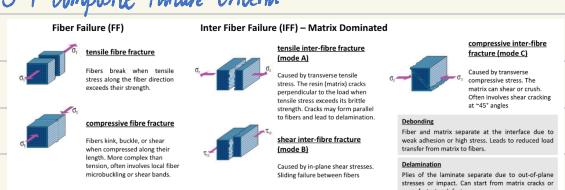
Antisymmetric Laminate

$[30^\circ/-30^\circ]$ $[30^\circ/-30^\circ/30^\circ/-30^\circ]$

$A_{16}=A_{26}=0$ $D_{16}=D_{26}=0$ no shear-tension / bending-twist coupling

$B=\begin{bmatrix} 0 & 0 & B_{16} \\ 0 & 0 & B_{26} \\ B_{16} & B_{26} & 0 \end{bmatrix}$ in plane tension-twist / shear-bending coupling

3.4 Composite Failure Criteria



Maximum Stress Criterion: Each stress component is compared independently to its strength. Failure occurs if any single stress exceeds the corresponding allowable.

Fiber direction: $\sigma_1 \geq X_t$ → fiber tension failure $\sigma_1 \leq -X_c$ → fiber compression failure

Transverse direction: $\sigma_2 \geq Y_t$ → matrix tension failure $\sigma_1 \leq -Y_c$ → matrix compression failure

Shear: $|\tau_{12}| \geq S$ → shear failure

+ simple to use. **- no interaction between stress.**
+ provides quick conservative estimates. **- overly conservative for multiaxial loading.**

Maximum Strain Criterion: Each strain component is compared independently to its strain limits. Failure occurs if any strain limit is exceeded.

Fiber direction: $\epsilon_1 \geq \epsilon_{1t}$ → fiber tension failure $\epsilon_1 \leq -\epsilon_{1c}$ → fiber compression failure

Transverse direction: $\epsilon_2 \geq e_{2t}$ → matrix tension failure $\epsilon_2 \leq -\epsilon_{2c}$ → matrix compression failure

Shear: $|\gamma_{12}| \geq \gamma_{12}^{max}$ → shear failure

+ sometimes more reliable than stress as strain data easier to measure. **- no interaction between strains/stresses.**
+ simple to apply. **- overly conservative for multiaxial loading.**

Tsai-Wu Criterion: Tsai–Wu is an interactive quadratic criterion considering stress interactions and distinguishes between tension and compression.

$$F_1\sigma_1 + F_2\sigma_2 + F_{11}{\sigma_1}^2 + F_{22}{\sigma_2}^2 + F_{66}{\tau_{12}}^2 + 2F_{12}\sigma_1\sigma_2 \geq 1$$

$$F_1 = \frac{1}{X_t} - \frac{1}{X_c} \qquad F_2 = \frac{1}{Y_t} - \frac{1}{Y_c} \qquad F_{12} = = \frac{1}{2}\sqrt{F_{11}F_{22}}$$

$$F_{11} = \frac{1}{X_t X_c} \qquad F_{22} = \frac{1}{Y_t Y_c} \qquad F_{66} = \frac{1}{S^2}$$

+ Captures interaction between stresses
+ Distinguishes between tension and compression

- no information about which failure mode occurs → just failure.

Hashin Criterion: The Hashin criterion is a set of failure conditions that distinguish between different modes of failure in fiber-reinforced composites. Instead of using a single overall strength criterion, it separately considers fiber and matrix failures in tension and compression.

- **Fiber tension failure $\sigma_1 \geq 0$:**

$$\left(\frac{\sigma_{11}}{X_t}\right)^2 + \frac{\tau_{12}^2}{S^2} = 1$$

- **Compressive failure $\sigma_1 \leq 0$:** fibers buckle or crush

$$\frac{|\sigma_{11}|}{X_c} = 1$$

- **Matrix tension failure $\sigma_2 \geq 0$:** matrix cracking transverse to fiber

$$\left(\frac{\sigma_{22}}{Y_t}\right)^2 + \frac{\tau_{12}^2}{S^2} = 1$$

- **Matrix compression failure $\sigma_2 \leq 0$:** matrix crushing due to shear failure

$$\frac{|\sigma_{22}|}{2S} + \left(\left(\frac{\tau_{12}}{S}\right)^2 = 1\right.$$

+ Mode-Specific Failure, separates between fibre vs matrix and tensile vs compressive failure

+ Widely accepted in aerospace and automotive composite modelling

- Only for Unidirectional Laminas. Developed mainly for UD composites. Less accurate for woven fabrics, multidirectional laminates, or hybrid composites.

- Limited to In-Plane Stresses

- **Puck Failure Criterion:**
  - distinguishes between fibre failure (FF) and inter-fibre failure (IFF)
  - handles different crack angles in the matrix, not just tension/compression.
  - More accurate for matrix-dominated failures.
  - More complex math needs extra material characterisation and input data.

- **LaRC03 / LaRC05 (NASA Models):**
  - Developed by NASA (Dávila & Camanho). Considered for polymer matrix composites.
  - Accounts for nonlinear shear behavior, fiber kinking, and fracture mechanics concepts.
  - Can predict initiation + propagation of different failures.
  - Needs detailed material characterisation.

Failure Mode	Max Stress / Max Strain	Tsai–Hill	Tsai–Wu	Hashin	Puck	Notes / Comments
Fiber-Dominated	✅ Captures basic fiber tension & compression	✅ Captures interaction, but no tension/compression distinction	✅ Captures fiber, tension vs compression included	✅ Explicit fiber tension & compression	✅ Explicit fiber tension & compression	All criteria predict fiber failure reasonably well, but Puck & Hashin are mode-specific
Matrix-Dominated / IFF	✅ Basic check for transverse stress or shear	❌ Not directly predicted	✅ Captures interaction, tension vs compression	✅ Distinguishes matrix tension/compression & shear	✅ Explicit inter-fiber fracture (tension, compression, shear)	Puck is most detailed; Hashin also widely used; Max Stress/Strain crude
Delamination / Interlaminar	❌ Not captured	❌ Not captured	❌ Not captured	❌ Not captured	❌ Not captured	Requires **fracture mechanics / cohesive zone modeling**; none of the ply-level criteria handle delamination directly

- **Max Stress / Max Strain** → simplest, first-pass checks, conservative.
- Tsai–Hill → stress interaction included, but no tension/compression distinction; mostly fiber failure.
- **Tsai–Wu** → interaction + tension/compression distinction; still phenomenological.
- Hashin → distinguishes fiber vs matrix modes, tension vs compression, widely used in FE.
- **Puck** → most advanced for practical composite laminate design, especially inter-fiber fracture, but more complex.
- Delamination → never captured by ply-level criteria; always needs 3D fracture mechanics.

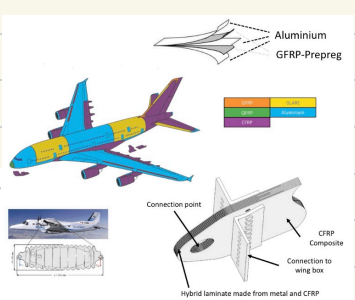
4. Hybrid Material

GLARE (Glass Laminate Aluminium Reinforced Epoxy) made from aluminium and GFRP

10% lighter than aluminium slightly reduced Young's Modulus

very good impact behaviour Higher fabrication effort

Good fatigue resistance



Application Example: Load connection points

- Heavily stressed material at load connection points e.g. bearing pressure at rivet- and bolt connections/joiner. Metal alloys better applicable.

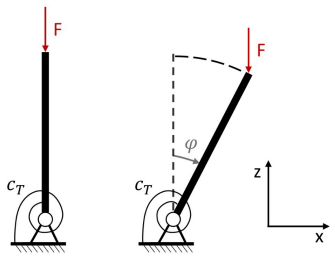
V. Elasto Stability

Problem: when elastic structures take compression or shear, sometimes they fail or deform when the load doesn't reach limitation

Strength: whether material can take load without bucking or fracture

Stability: whether the structure remain or restore to initial position after slight distribution

2. Rigid Column under Compression Load



Assumptions:

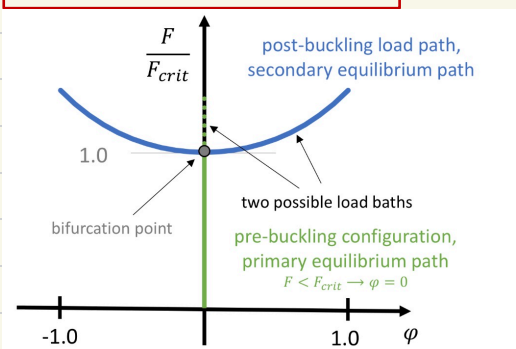
- The force *F* maintains its direction
- The elastic spring has a constant torsional stiffness C_T.

Equilibrium Equation: $F \cdot \sin\varphi \cdot l - \varphi \cdot C_T = 0$

$[Fl(1-\frac{\varphi^2}{6}) - C_T] \cdot \varphi = 0 \Rightarrow$ stability condition

case 1: $\varphi = 0$

case 2: $F(\varphi) = (\frac{6}{6-\varphi^2}) F_{crit}$ $\quad F_{crit} = \frac{C_T}{l}$



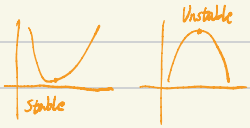
- Pre-bucking: Compression only, stable when $F < F_{crit}$
- when F reaches F_{crit} and keeps grow, there will be two possible path (从这一刻起，历史发生了巨变)
- $F > F_{crit}$, column keep straight, but lose stability, any slight disturbance will cause change But position changing doesn't mean collapse or fracture, if it go to blue curve will be stable again.
- $\varphi \neq 0$, if F satisfy $F(\varphi) = F_{crit}(1-\frac{\varphi^2}{6})$ will be stable
  if not, position change, further discussion

2.2 Energy Method

Potential Energy (spring): $\Pi_i = \frac{1}{2} C_T \varphi^2$ External Work: $W_e = \int -F\,dz = Fl(1-\cos\varphi)$

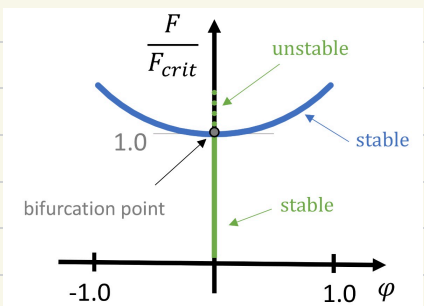
Potential Energy $\Pi = -W_e + \Pi_i = Fl(1-\cos\varphi) + \frac{1}{2} C_b \varphi^2$

Static Equilibrium: $\frac{d\Pi}{d\varphi} = -Fl\sin\varphi + C_b\varphi = 0$ $Fl\sin\varphi = C_b\varphi$ (min or max)



second derivative is needed: $\frac{d^2\Pi}{d\varphi^2} = -Fl\cos\varphi + C_T$

if $\frac{d^2\Pi}{d\varphi^2} > 0$, minimum, stable
if $\frac{d^2\Pi}{d\varphi^2} < 0$, maximum, unstable



Pre-buckling: $\varphi = 0$

$\frac{d^2\Pi}{d\varphi^2}(\varphi=0) = -Fl + C_T$

$F < F_{crit}, F < \frac{C_T}{l}$, stable

$F > F_{crit}, F > \frac{C_T}{l}$, unstable

Post buckling: $\varphi \neq 0$, $F > F_{crit}$, $F > \frac{C_T}{l}$

Taylor series: $\cos\varphi = 1 - \frac{\varphi^2}{2} + O(\varphi^4)$ $\sin\varphi = \varphi - \frac{\varphi^3}{6} + O(\varphi^5)$

$\frac{d^2\Pi}{d\varphi^2} = -Fl(1-\frac{\varphi^2}{2}) + C_T$

$\varphi^2 = 6 - \frac{6C_T}{Fl}$

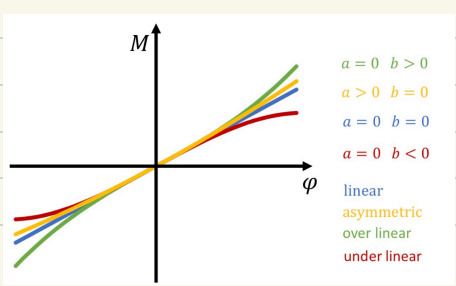
$\frac{d^2\Pi}{d\varphi^2} = -Fl(1 - \frac{6-\frac{6C_T}{Fl}}{2}) + C_T = 2Fl + 4C_T > 0$ stable

2.4 Influence of Structure Nonlinearities

$M(\varphi) = C_T\varphi(1 + a\varphi + b\varphi^2)$ $\Pi_i = \int M(\varphi)\,d\varphi = C_T(\frac{\varphi^2}{2} + a\frac{\varphi^3}{2} + b\frac{\varphi^4}{4})$ $\Pi = \Pi_i - W_e$

Equilibrium Equation: $\frac{\partial\Pi}{\partial\varphi} = C_T\varphi(1 + \frac{3}{2}a\varphi + b\varphi^2) - Fl\sin\varphi = 0$

1) $\varphi = 0$
2) $F = F_{crit}\frac{\varphi + a\varphi^2 + b\varphi^3}{\sin\varphi}$



Stability: $\Pi'' = C_T(1+3a\varphi+3b\varphi^2) - F\cos\varphi$

① $\varphi = 0$

$\Pi''(0) = C_T - Fl$ $F_{crit} = \frac{C_T}{l}$

$\frac{F}{F_{crit}} < 1$, $\Pi''(0) > 0$, stable

$\frac{F}{F_{crit}} > 1$, $\Pi''(0) < 0$, unstable

② $a = 0, b = 0$ — linear spring

$\Pi'' = C_T - F\cos\varphi$

Pre buckling: $\varphi = 0$ Post buckling: $\varphi \neq 0$

$F_{crit} = \frac{C_T}{l}$ $F = \left(\frac{6}{6-\varphi^2}\right) F_{crit}$

$\frac{F}{F_{crit}} < 1$, stable

$\frac{F}{F_{crit}} > 1$, unstable

③ $a = 0, b = 1$ — harden spring

$\Pi'' = C_T(1+3\varphi^2) - F\cos\varphi$

$\frac{F}{F_{crit}} = \frac{\varphi+\varphi^3}{\sin\varphi} > 1$

$\Pi'' \approx C_T(1+3\varphi^2 - \cos\varphi) \approx C_T\left(1+3\varphi^2 - \left(1-\frac{\varphi^2}{2}\right)\right) = \frac{7}{2}C_T\varphi^2 > 0$

Stable

④ $a = 0, b = -1$ — soften spring

$\Pi'' = C_T(1-3\varphi^2) - F\cos\varphi$

$\frac{F}{F_{crit}} = \frac{\varphi-\varphi^3}{\sin\varphi} < 1$

$\Pi'' \approx C_T\left(1-3\varphi^2 - \left(1-\frac{\varphi^2}{2}\right)\right) = -\frac{5}{2}C_T\varphi^2 < 0$

Unstable

⑤ $a = 1, b = 0$

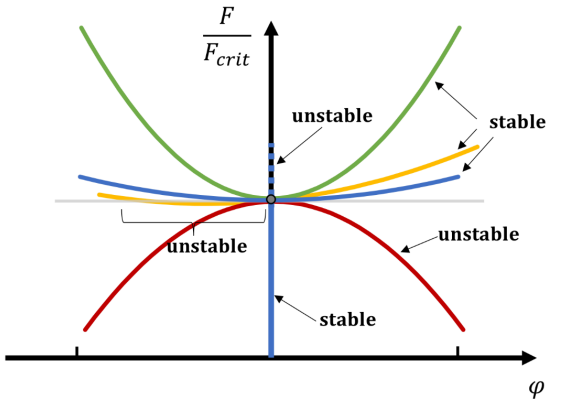
$\Pi'' = C_T(1+3\varphi) - F\cos\varphi$

$\Pi'' \approx C_T\left(1+3\varphi - \left(1-\frac{\varphi^2}{2}\right)\right) = C_T\left(3\varphi + \frac{\varphi^2}{2}\right) = C_T\varphi\left(3+\frac{\varphi}{2}\right)$

$\varphi > 0$: $\Pi'' > 0$

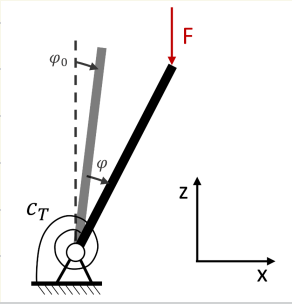
$\varphi < 0$: $\left(3+\frac{\varphi}{2}\right) > 0 \Rightarrow \Pi'' < 0$

$\left(3+\frac{\varphi}{2}\right) < 0 \Rightarrow \Pi'' > 0$



- The load-deformation curves in nonlinear elastic systems can be very different compared to linear systems., resulting in different stability characteristics.

2.5 Influence of Imperfections



With initial deflection φ_0

Spring Stiffness : $M(\varphi) = c_T \cdot \varphi(1 + a\varphi + b\varphi^2)$

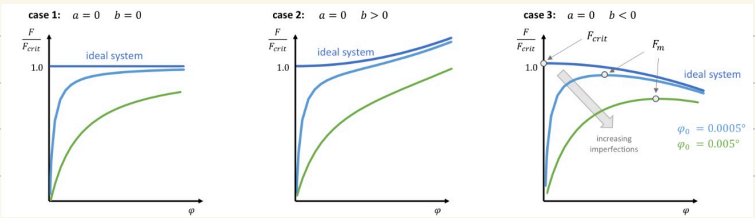
External Work : $W_e = -\int F dz = F \cdot l(\cos\varphi_0 - \cos(\varphi_0 + \varphi))$

Potential Energy : $\pi = -W_e + \pi_i = c_T(\frac{\varphi^2}{2} + a\frac{\varphi^3}{3} + b\frac{\varphi^4}{4}) - Fl(\cos\varphi_0 - \cos(\varphi_0 + \varphi))$

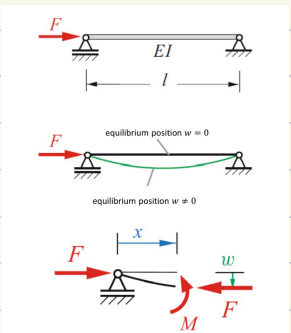
Static Equilibrium : $\frac{\partial \pi}{\partial \varphi} = c_T(\varphi + a\varphi^2 + b\varphi^3) - Fl\sin(\varphi_0 + \varphi)$

$$\frac{F}{F_{crit}} = \frac{\varphi + a\varphi^2 + b\varphi^3}{\sin(\varphi_0 + \varphi)}$$

consider $\varphi_0 = 0.0005°$ and $\varphi_0 = 0.05°$



3.1 Elastic Column under Compression Load



Arbitrary boundary conditions need to include the transverse loading Q. the buckling DGL becomes:

$$EIw'''' + F \cdot w'' = 0$$

Euler-Bernoulli Beam : $w'' = -\frac{M}{EI}$ Static Equilibrium : $M = F \cdot w$

Buckling Differential Equation : $-EIw'' = Fw \Rightarrow w'' + \frac{F}{EI}w = 0$

$\lambda^2 = \frac{F}{EI}\,(m^{-1})$, $w'' + \lambda^2 w = 0$

Solution : $w(x) = e^{rx}$, $w'' = r^2 e^{rx}$ $r^2 e^{rx} + \lambda^2 e^{rx} = 0 \Rightarrow r^2 + \lambda^2 = 0$, $r = \pm i\lambda$

$w(x) = C_1\cos(\lambda x) + C_2\sin(\lambda x)$

BC's : $w(0) = 0$, $w(l) = 0$

$w(0) = C_1 = 0$, $w(x) = C_2\sin(\lambda x)$

$w(l) = C_2\sin(\lambda l) = 0$

1) $C_2 = 0$, $w(x) = 0$ - straight, trivial solution

2) $\sin(\lambda l) = 0$, $\lambda l = n\pi$, $\lambda_n = \frac{n\pi}{l}$

Critical Buckling load : $\lambda^2 = \frac{F}{EI}$, $F = \lambda^2 EI$, $F_n = \frac{n^2\pi^2 EI}{l^2}$

$$\lambda_1 = \frac{\pi}{l}, \quad F_{crit} = \pi^2\frac{EI}{l^2}$$

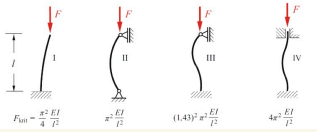
With transverse load Q

$EIw'''' + F \cdot w'' = 0$

$w'''' + \lambda^2 \cdot w'' = 0$

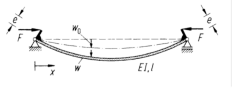
$w(x) = C_1\cos(\lambda x) + C_2\sin(\lambda x) + C_3 x + C_4$

Euler Buckling Cases:



3.1 Imperfect Elastic Column under Compression Load

Like the rigid body beam, as soon as imperfections are present, deflections occur even with small loads.



Static equilibrium: $M = F \cdot (w_0 + e + w)$

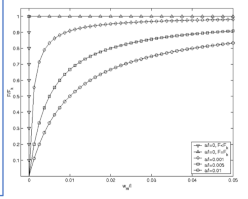
Buckling DGL: $\mathbf{EI}\, \boldsymbol{w''} + \boldsymbol{Fw} = -F \cdot (w_0 + e)$

Initial deformation: $w_0 = a_0 \cdot \sin\left(\frac{\pi x}{l}\right)$

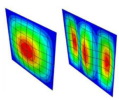
Approach: $w = A\cos(\lambda x) + B\sin(\lambda x) + \frac{a_0}{(\pi/\lambda l)^2 - 1} sin\left(\frac{\pi x}{l}\right) - e$

solution: $w = \frac{a_0}{(\pi/\lambda l)^2 - 1} sin\left(\frac{\pi x}{l}\right) + e\left(\frac{\cos\left(\frac{\lambda l}{2} - \lambda x\right)}{\cos\left(\frac{\lambda l}{2}\right)} - 1\right)$

$w_{max} = \frac{a_0}{(\pi/\lambda l)^2 - 1} + e\left(\frac{1}{cos\left(\frac{\lambda l}{2}\right)} - 1\right) = \frac{a_0}{\frac{F_{crit}}{F} - 1} + e\left(\frac{1}{\cos\left(\frac{\pi}{2}\sqrt{F/F_{crit}}\right)} - 1\right)$

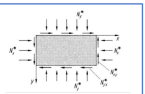


3.2 Elastic Plate under Compression Load



Like the beam, a plane surface structure can also deflect laterally under compression or shear loads acting in the plane alone. This is referred to as plate buckling.

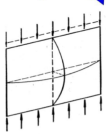
Normal and Shear Forces:



$N_x = -\int \sigma_x dz = -\sigma_x t$ $\quad N_y = -\int \sigma_y dz = -\sigma_y t$

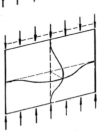
$N_{xy} = -\int \tau_{xy} dz = -\tau_{xy} t$ $\quad N_{yx} = -\int \tau_{yx} dz = -\tau_{yx} t$

Equilibrium Condition: $\frac{\partial N_x}{\partial x} + \frac{\partial N_{yx}}{\partial y} = 0$ $\quad \frac{\partial N_y}{\partial y} + \frac{\partial N_{xy}}{\partial y} = 0$



Equilibrium Condition: (deformed system)

$\frac{\partial^2 M_x}{\partial x^2} + 2\frac{\partial^2 M_{xy}}{\partial x \partial y} + \frac{\partial^2 M_y}{\partial y^2} - N_x \frac{\partial^2 w}{\partial x^2} - 2N_{xy}\frac{\partial^2 w}{\partial x \partial y} - N_y \frac{\partial^2 w}{\partial y^2} = 0$

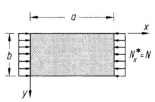


Isotropic constitutive law (Hook's law), assuming Kirchhoff Plate Theory:

$$K\left(\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial x^4}\right) + N_x \frac{\partial^2 w}{\partial x^2} + 2N_{xy}\frac{\partial^2 w}{\partial x \partial y} + N_y \frac{\partial^2 w}{\partial y^2} = 0$$

3.2 Elastic Plate under Compression Load

A solution in closed form can be found only in special cases. As an example, we will consider the plate loosely supported on all sides under constant pressure load N on one side. It follows in $N_x = N, N_y = N_{xy} = 0$.



Buckling DGL: $K\left(\frac{\partial^4 w}{\partial x^4} + 2\frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial x^4}\right) + N\frac{\partial^2 w}{\partial x^2} = 0$

Edge Ratio: $\alpha = \frac{a}{b}$

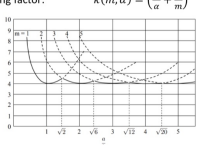
Fourier Approach: $w(x,y) = \Sigma_m \Sigma_n w_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi y}{b}\right) \quad m, n = 1, 2, 3 \ldots$

$N = \frac{\pi^2 K}{b^2}\left(\frac{m}{\alpha} + \frac{\alpha}{m}\right)^2 = k(m,\alpha)\frac{\pi^2 K}{b^2}$

BC's: $w(0,y) = w(a,y) = 0 \quad w(x,0) = w(x,b) = 0$

$M_x(0,y) = M_x(a,y) = 0 \quad M_y(x,0) = M_y(x,b) = 0$

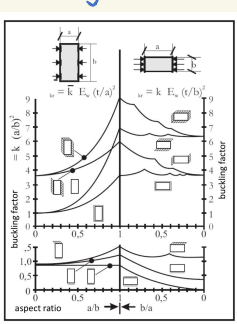
Buckling factor: $k(m,\alpha) = \left(\frac{m}{\alpha} + \frac{\alpha}{m}\right)^2$



$\Sigma_m \Sigma_n \left(K\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]^2 - N\left(\frac{m\pi}{a}\right)^2\right) w_{mn} \sin\left(\frac{m\pi x}{a}\right) \sin\left(\frac{n\pi x}{b}\right) = 0$

$$N = \frac{K\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right]^2}{\left(\frac{m\pi}{a}\right)^2}$$

4. Buckling Tables



Horizontal: aspect ratio ($\frac{a}{b} \leq 1$ and $\frac{b}{a} \leq 1$)

Vertical: buckling factor $\bar{K}/k$ ($\frac{\bar{K}}{K} = \left(\frac{a}{b}\right)^2$)

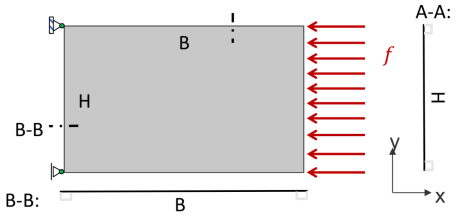
Different BC's

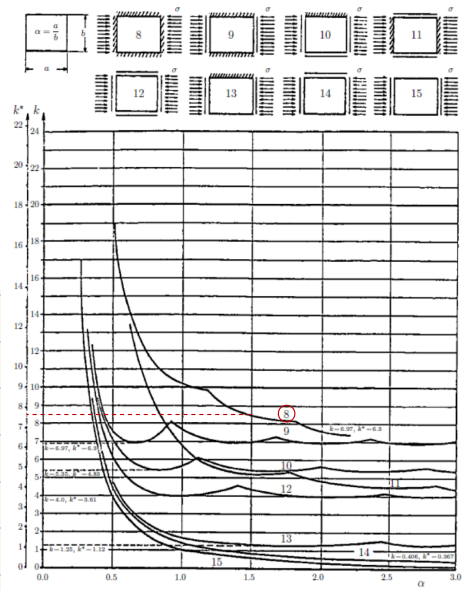
4. Buckling Tables - Compression

Panel Details:

H = 400 mm f = 40 N/m
B = 600 mm

Panel thickness: t = 0,15 mm
Material: St (E = 210000 MPa, G = 70000 MPa, v = 0,3)





$N_{crit} = K\frac{\pi^2 K}{b^2}$

$K = \frac{Et^3}{12(1-\nu^2)}$

$N_{crit} = K\frac{\pi^2}{b^2}\frac{Et^3}{12(1-\nu^2)}$

$\frac{N}{b} = K\frac{\pi^2}{b^2}\frac{Et^3}{12(1-\nu^2)}\frac{1}{t}$

$\sigma_{cr} = K\frac{\pi^2 E}{12(1-\nu^2)}\frac{t^2}{b^2} = K^* \cdot E \cdot \left(\frac{t}{b}\right)^2 \quad , K^* = K\frac{\pi^2}{12(1-\nu^2)}$

$\alpha = \frac{a}{b} = 1.5$

$K^* = 7.8$

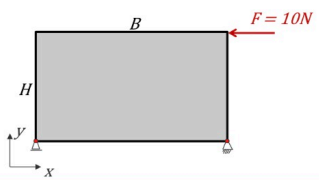
$\sigma_{cr} = K^* \cdot E \cdot \left(\frac{t}{H}\right)^2 = 0.23\,MPa$

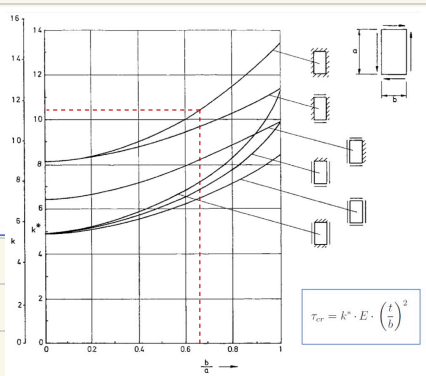
$\sigma = \frac{F}{A} = \frac{t \cdot H}{H \cdot b} = 0.6\,MPa$

$\sigma > \sigma_{cr}$

4.2 Buckling Tables - Shear

t=0.15mm
B=600mm
H=400mm





$\alpha = 0.66$

$K^* = 10.5$

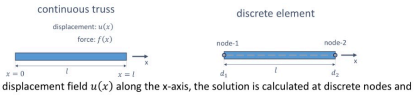
$\tau_{cr} = K^* \cdot E \cdot \left(\frac{t}{b}\right)^2 = 0.316\,MPa$

$\tau = \frac{F}{A} = 0.111\,MPa < \tau_{cr}$

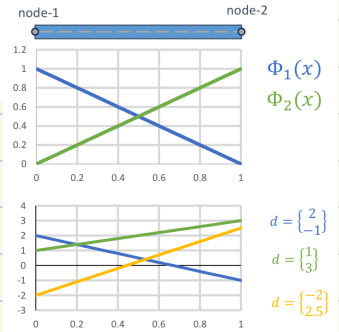
VI. FEM

Target: Transform partial differential equations (PDE) into differential equations

Example 1: Truss Element



Idea: Instead of solving the displacement field $u(x)$ along the x-axis, the solution is calculated at discrete nodes and interpolated in-between.



Assumption linear displacement field: $V(x)=a_0+a_1x$

BC's: $V(0)=d_1$ $V(l)=d_2 \Rightarrow a_0=d_1$, $a_1=\frac{d_2-d_1}{l}$

$V(x)=d_1+(\frac{d_2-d_1}{l})x=(1-\frac{x}{l})d_1+(\frac{x}{l})d_2$

$V(x)=[(1-\frac{x}{l})\ (\frac{x}{l})]\begin{bmatrix}d_1\\d_2\end{bmatrix}=\underline{\Phi}^T\cdot\vec{d}$

$\underline{\Phi}=\begin{bmatrix}\underline{\Phi}_1(x)\\\underline{\Phi}_2(x)\end{bmatrix}=\begin{bmatrix}1-\frac{x}{l}\\\frac{x}{l}\end{bmatrix}$ Shape Functions (vary with positions)

Contributions to the displacement of a certain point of displacement of different points

$\underline{\Phi}_1$: contribution to the displacement filed of displacement of dot 1 d_1

$\underline{\Phi}_2$: contribution to the displacement filed of displacement of dot 1 d_2

Required: shape function $\underline{\Phi}_n=1$ at dot n and 0 at other dots

Displacement Field: $V=\underline{\Phi}^T\cdot\vec{d}$

Velocity Field: $\dot{V}=\underline{\Phi}^T\cdot\dot{\vec{d}}$

Strain: $U_{ix}=\underline{\Phi}'^T_x\cdot\vec{d}$

Strain Energy: $V=\int_V\sigma\epsilon\,dV=\frac{1}{2}\int_0^l EAV_x'\,dx=\frac{1}{2}EA\int_0^l U_x'\,dx=\frac{1}{2}\vec{d}^T\,EA\overbrace{\int_0^l\underline{\Phi}'_x\cdot\underline{\Phi}'^T_x\,dx}^{[K]}\cdot\vec{d}$

$V=\frac{1}{2}\vec{d}^T[K]\vec{d}$

Stiffness Matrix: $[K]=\int_0^l\underline{\Phi}'_x\cdot\underline{\Phi}'^T_x\,dx=EA\cdot\int_0^l\begin{bmatrix}\frac{1}{l}\\-\frac{1}{l}\end{bmatrix}[-\frac{1}{l}\ \frac{1}{l}]\,dx=EA\int_0^l\begin{bmatrix}\frac{1}{l^2}&-\frac{1}{l^2}\\-\frac{1}{l^2}&\frac{1}{l^2}\end{bmatrix}dx=EA\begin{bmatrix}\frac{1}{l}&-\frac{1}{l}\\-\frac{1}{l}&\frac{1}{l}\end{bmatrix}$

$[K]=\frac{EA}{l}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}$

Virtual Work: $\delta W=\int_0^l f(x)\cdot\delta u\,dx=\int_0^l\overbrace{f(x)\cdot\underline{\Phi}^T}^{[F]}\cdot\vec{d}\,dx$

$\frac{d}{dt}(\frac{\partial T}{\partial\dot{q}_i})-\frac{\partial T}{\partial q_i}+\frac{\partial V}{\partial q_i}=Q_i$

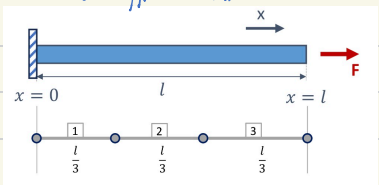
$V=\frac{1}{2}\vec{d}[K]\vec{d}$ $\frac{\partial V}{\partial\vec{d}}=[K]\vec{d}$

$\delta W=\vec{F}\cdot\vec{d}$ $\frac{\partial W}{\partial\vec{d}}=\vec{F}$

Load Vector: $\vec{F}=[K]\vec{d}$

$\begin{bmatrix}f_1\\f_2\end{bmatrix}=\frac{EA}{l}\begin{bmatrix}1&-1\\-1&1\end{bmatrix}\begin{bmatrix}d_1\\d_2\end{bmatrix}$

3. Global Stiffness Matrix



$[K]_1 = \frac{EA}{l_e}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ (1,2) $\quad [K]_2 = \frac{EA}{l_e}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ (2,3) $\quad [K]_3 = \frac{EA}{l_e}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$ (3,4)

Global Stiffness Matrix:

$$\frac{EA}{l_e}\begin{bmatrix}1 & -1 & 0 & 0\\ -1 & 1+1 & -1 & 0\\ 0 & -1 & 1+1 & -1\\ 0 & 0 & -1 & 1\end{bmatrix}\begin{bmatrix}d_1\\ d_2\\ d_3\\ d_4\end{bmatrix} = \begin{bmatrix}f_1\\ f_2\\ f_3\\ f_4\end{bmatrix}$$

BC's: displacement $d_1 = 0$ Forces: $f_4 = F$ $f_2 = f_3 = 0$

$$\frac{EA}{l_e}\begin{bmatrix}1 & 1 & 0 & 0\\ 1 & 2 & -1 & 0\\ 0 & -1 & 2 & -1\\ 0 & 0 & -1 & 1\end{bmatrix}\begin{bmatrix}0\\ d_2\\ d_3\\ d_4\end{bmatrix} = \begin{bmatrix}f_1\\ 0\\ 0\\ F\end{bmatrix}$$

f_x / u_x : known

f_x / u_x : unknown

$$\frac{EA}{l_e}\begin{bmatrix}2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 1\end{bmatrix}\begin{bmatrix}d_2\\ d_3\\ d_4\end{bmatrix} = \begin{bmatrix}0\\ 0\\ F\end{bmatrix}$$

$$\begin{bmatrix}d_1\\ d_2\\ d_3\end{bmatrix} = \frac{l_e}{EA}\begin{bmatrix}2 & -1 & 0\\ -1 & 2 & -1\\ 0 & -1 & 1\end{bmatrix}^{-1}\cdot\begin{bmatrix}0\\ 0\\ F\end{bmatrix} = \frac{Fl_e}{EA}\begin{bmatrix}1\\ 2\\ 3\end{bmatrix}$$

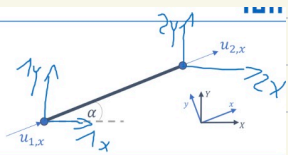
Displacement: $v(x) = \begin{bmatrix}(1-\frac{x}{l}) & (\frac{x}{l})\end{bmatrix}\begin{bmatrix}d_1\\ d_2\end{bmatrix}$

Strains: $\varepsilon = \frac{du}{dx} = \begin{bmatrix}-\frac{1}{l} & \frac{1}{l}\end{bmatrix}\begin{bmatrix}d_1\\ d_2\end{bmatrix} = \frac{d_2 - d_1}{l}$ $\quad \varepsilon = \frac{1}{l}\begin{bmatrix}-c & -s & c & s\end{bmatrix}\begin{bmatrix}u_{1x}\\ u_{1y}\\ u_{2x}\\ u_{2y}\end{bmatrix}$

Stress: $\sigma = E\cdot\varepsilon = \frac{E}{l}\begin{bmatrix}-1 & 1\end{bmatrix}\begin{bmatrix}d_1\\ d_2\end{bmatrix}$ $\quad \sigma = E\cdot\varepsilon = \frac{E}{l}\begin{bmatrix}-c & -s & c & s\end{bmatrix}\begin{bmatrix}u_{1x}\\ u_{1y}\\ u_{2x}\\ u_{2y}\end{bmatrix}$

3. Local vs Global Coordinate System

In the finite element method, a distinction is made between so-called **local** or **element** coordinates and **global** coordinates.



Bar Stiffness Matrix $k_L = \frac{EA}{l}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}$

Rotation Matrix

$$\begin{bmatrix}u_{1x}\\ u_{1x}\end{bmatrix} = \begin{bmatrix}\cos\alpha & \sin\alpha & 0 & 0\\ 0 & 0 & \cos\alpha & \sin\alpha\end{bmatrix}\begin{bmatrix}u_{1x}\\ u_{1x}\\ u_{1y}\\ u_{2y}\end{bmatrix}$$

$\vec{d}_i$ $\quad [L]$ $\quad \vec{D}_i$

Stiffness Matrix Global Coordinates

$K_e = [L]^T[K_L][L]$

$$K_e = \begin{bmatrix}\cos\alpha & 0\\ \sin\alpha & 0\\ 0 & \sin\alpha\\ 0 & \cos\alpha\end{bmatrix}\frac{EA}{l}\begin{bmatrix}1 & -1\\ -1 & 1\end{bmatrix}\begin{bmatrix}\cos\alpha & \sin\alpha & 0 & 0\\ 0 & 0 & \cos\alpha & \sin\alpha\end{bmatrix}$$

$$K_e = \frac{EA}{l}\begin{bmatrix}c^2 & sc & -c^2 & -sc\\ sc & s^2 & -sc & -s^2\\ -c^2 & -sc & c^2 & sc\\ -sc & -s^2 & sc & s^2\end{bmatrix}$$

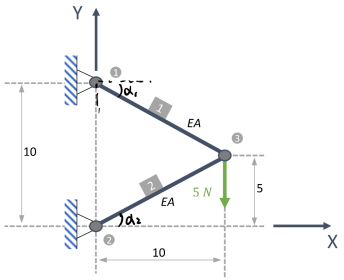
s: sin

c: cos

$V = \frac{1}{2}\vec{d}_i[K_e]\vec{d}_i = \frac{1}{2}\vec{D}_i[L]^T[K_e][L]\vec{D}_i$

3. Truss Structure Modelling

Task 1: Find the displacement of node 3.
Task 2: Calculate the stresses in the element.



A := constant cross section area
E := Youngs' modulus

Element 1: $\alpha_1 = -\arctan(\frac{5}{10}) = -26.57°$

$s = \sin(-26.57°) = -0.42 \quad c = \cos(-25.57°) = 0.91$

$$K_{e1} = \begin{bmatrix} s^2 & sc & -s^2 & -sc \\ sc & c^2 & -sc & -c^2 \\ -s^2 & -sc & s^2 & sc \\ -sc & -c^2 & sc & c^2 \end{bmatrix} = \begin{bmatrix} 0.8 & -0.4 & -0.8 & 0.4 \\ -0.4 & 0.18 & 0.4 & -0.18 \\ -0.8 & 0.4 & 0.8 & -0.4 \\ 0.4 & -0.18 & -0.4 & 0.18 \end{bmatrix} \begin{matrix} 1x \\ 1y \\ 3x \\ 3y \end{matrix}$$

Element 2: $\alpha_2 = 25.57° \quad s = 0.42 \quad c = 0.91$

$$K_{e2} = \begin{bmatrix} 0.8 & 0.4 & -0.8 & -0.4 \\ 0.4 & 0.18 & -0.4 & -0.18 \\ -0.8 & -0.4 & 0.8 & 0.4 \\ -0.4 & -0.18 & 0.4 & 0.18 \end{bmatrix} \begin{matrix} 2x \\ 2y \\ 3x \\ 3y \end{matrix}$$

$$K_e = \begin{bmatrix} 0.8 & -0.4 & 0 & 0 & -0.8 & 0.4 \\ -0.4 & 0.18 & 0 & 0 & 0.4 & -0.18 \\ 0 & 0 & 0.8 & 0.4 & -0.8 & -0.4 \\ 0 & 0 & 0.4 & 0.18 & -0.4 & -0.18 \\ -0.8 & 0.4 & -0.8 & -0.4 & 1.6 & 0 \\ 0.4 & -0.18 & -0.4 & -0.18 & 0 & 0.36 \end{bmatrix} \begin{matrix} 1x \\ 1y \\ 2x \\ 2y \\ 3x \\ 3y \end{matrix}$$

$$\frac{EA}{l} \begin{bmatrix} 0.8 & -0.4 & 0 & 0 & -0.8 & 0.4 \\ -0.4 & 0.18 & 0 & 0 & 0.4 & -0.18 \\ 0 & 0 & 0.8 & 0.4 & -0.8 & -0.4 \\ 0 & 0 & 0.4 & 0.18 & -0.4 & -0.18 \\ -0.8 & 0.4 & -0.8 & -0.4 & 1.6 & 0 \\ 0.4 & -0.18 & -0.4 & -0.18 & 0 & 0.36 \end{bmatrix} \begin{bmatrix} D_{1x} \\ D_{1y} \\ D_{2x} \\ D_{2y} \\ D_{3x} \\ D_{3y} \end{bmatrix} = \begin{bmatrix} F_{1x} \\ F_{1y} \\ F_{2x} \\ F_{2y} \\ F_{3x} \to 0 \\ F_{3y} \to -F \end{bmatrix}$$

$$\frac{EA}{l} \begin{bmatrix} 1.6 & 0 \\ 0 & 0.36 \end{bmatrix} \begin{bmatrix} D_{3x} \\ D_{3y} \end{bmatrix} = \begin{bmatrix} 0 \\ -F \end{bmatrix}$$

$$\begin{bmatrix} D_{3x} \\ D_{3y} \end{bmatrix} = \frac{Fl}{EA} \begin{bmatrix} 1.6 & 0 \\ 0 & 0.36 \end{bmatrix}^{-1} \begin{bmatrix} 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ -2.78\frac{Fl}{EA} \end{bmatrix}$$

stress: $\sigma = \frac{E}{l} [-1 \; 1] \begin{bmatrix} d_1 \\ d_2 \end{bmatrix}$

$$\begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} u_{1x} \\ u_{1y} \end{bmatrix} = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 & 0 \\ 0 & 0 & \cos\alpha & \sin\alpha \end{bmatrix} \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \end{bmatrix}$$

$$\sigma_1 = \frac{E}{l} [-1 \; 1] \begin{bmatrix} \cos\alpha_1 & \sin\alpha_1 & 0 & 0 \\ 0 & 0 & \cos\alpha_1 & \sin\alpha_1 \end{bmatrix} \begin{bmatrix} u_{1x} = 0 \\ u_{1y} = 0 \\ u_{2x} \\ u_{2y} \end{bmatrix}$$

$$\sigma_2 = \frac{E}{l} [-1 \; 1] \begin{bmatrix} \cos\alpha_2 & \sin\alpha_2 & 0 & 0 \\ 0 & 0 & \cos\alpha_2 & \sin\alpha_2 \end{bmatrix} \begin{bmatrix} u_{1x} = 0 \\ u_{1y} = 0 \\ u_{2x} \\ u_{2y} \end{bmatrix}$$

$$\sigma_1 = \frac{E}{l} [-1 \; 1] \begin{bmatrix} 0 & 0 \\ \cos\alpha_1 & \sin\alpha_1 \end{bmatrix} \begin{bmatrix} 0 \\ -2.78\frac{Fl}{EA} \end{bmatrix}$$

$$= 1.17\frac{F}{A}$$

$$\sigma_2 = \frac{E}{l} [-1 \; 1] \begin{bmatrix} 0 & 0 \\ \cos\alpha_2 & \sin\alpha_2 \end{bmatrix} \begin{bmatrix} 0 \\ -2.78\frac{Fl}{EA} \end{bmatrix} = -1.17\frac{F}{A}$$

3. Truss Structure Modelling - Summary

1. Problem Setup:
   - nodes
   - elements
   - DOF's
   - forces/displacement BC's
   - axes
   - material definitions
   - geometric properties

2. Problem Setup:
   - find element direction vector, calculate rotation matrix $[\tau]$
   - find element stiffness matrix $[k_e]$
   - Rotate element matrices to global coordinates $[k_r]$
   - Global stiffness matrix assembly $[K_G]$
   - Assemble force vector $\{F\}$

3. Remove Restrained DOF's

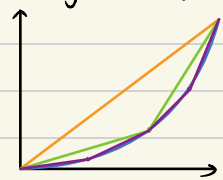
4. Solve/find Displacement:
   - $\{F\} = [K_G]\{D\}$

5. Solve strain and stresses:
   - Transform global to local coordinates $\{d\} = [\tau]\{D\}$
   - calculate strains $\varepsilon = \frac{\partial u}{\partial x}$
   - Calculate stress $\sigma = E \cdot \varepsilon$

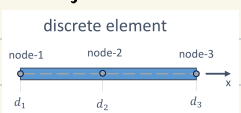
H- and P- Convergence

Target : increase accuracy

H-Convergence : increase number of Elements (refined grids)



P-Convergence : higher order shape functions



	$x_1=-1$	$x_2=0$	$x_3=1$
$N_1=\frac{x^2}{2}-\frac{x}{2}$	1	0	0
$N_2=(1+x)(1-x)$	0	1	0
$N_3=\frac{x^2}{2}+\frac{x}{2}$	0	0	1

4. Beam Element

Euler-Bernoulli Beam $\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} w_1 \\ \theta_1 \\ w_2 \\ \theta_2 \end{bmatrix}$ (do not consider tension)

Strain Energy : $U=\frac{1}{2}\int_0 \sigma\varepsilon dV = \frac{1}{2}\int_0^l EI[w''(x)]^2 dx$

$w(x)=[\vec{\phi}_1\ \vec{\phi}_2\ \vec{\phi}_3\ \vec{\phi}_4]\begin{bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \end{bmatrix}$

$U=\frac{1}{2}\vec{d}^T EI\int_0^l \vec{\phi}_{,xx}\cdot\vec{\phi}^T_{,xx}dx\cdot\vec{d} \longleftrightarrow U=\frac{1}{2}\vec{d}^T[K]\vec{d}$

$[k_e]=EI\int_0^l \vec{\phi}_{,xx}\cdot\vec{\phi}^T_{,xx}dx$

$\theta=\frac{dw}{dx}$

Displacement Field : $w(x)=a_0+a_1x+a_2x^2+a_3x^3$ BC's: $w_1=d_1$ $w_2=d_3$ $w_1'=d_2$ $w_2'=d_4$

$a_0=d_1$ $a_2=\frac{1}{l^2}(-3d_1-2d_2l+3d_3-d_4l)$

$a_1=d_2$ $a_3=\frac{1}{l^3}(2d_1+d_2l-2d_3+d_4l)$

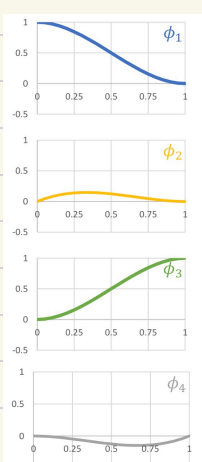
$\phi_1=1-3(\frac{x}{l})^2+2(\frac{x}{l})^3$ $\phi_2=x-2l(\frac{x}{l})^2+l(\frac{x}{l})^3$

$\phi_3=3(\frac{x}{l})^2-2(\frac{x}{l})^3$ $\phi_4=-l(\frac{x}{l})^2+l(\frac{x}{l})^3$

$w(x)=\phi_1d_1+\phi_2d_2+\phi_3d_3+\phi_4d_4=\vec{\phi}\cdot\vec{d}$

Stiffness Matrix : $[k_e]=EI\int_0^l \vec{\phi}_{,xx}\cdot\vec{\phi}^T_{,xx}dx=EI\int_0^l\begin{bmatrix}\phi_1'' \\ \phi_2'' \\ \phi_3'' \\ \phi_4''\end{bmatrix}\cdot\begin{bmatrix}\phi_1'' & \phi_2'' & \phi_3'' & \phi_4''\end{bmatrix}dx$

$[k_e]=\frac{EI}{l^3}\begin{bmatrix}12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2\end{bmatrix}$

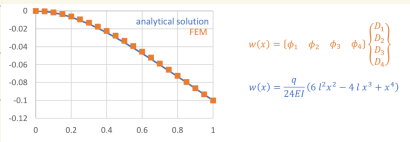


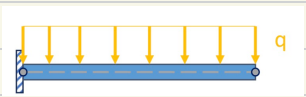
$$[k_e] = \frac{EI}{l^3}\begin{bmatrix} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{bmatrix}$$

$$\frac{EI}{l^3}\left[\begin{array}{cc|cc} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ \hline -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{array}\right]\begin{bmatrix} d_1 = 0 \\ d_2 \neq 0 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} F_1 \\ M_1 = 0 \\ F_2 \neq F \\ M_2 \neq 0 \end{bmatrix}$$

$$\frac{EI}{l^3}\begin{bmatrix} 12 & -6l \\ -6l & 4l^2 \end{bmatrix}\begin{bmatrix} d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} F \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} d_3 \\ d_4 \end{bmatrix} = \frac{Fl^3}{EI}\frac{1}{12\cdot 4l^2 - 36l^2}\begin{bmatrix} 4l^2 & 6l \\ 6l & 12 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{Fl^2}{12EI}\begin{bmatrix} 4l \\ 6 \end{bmatrix}$$





$$\begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix} = \int_0^l q \begin{bmatrix} 1-3\left(\frac{x}{l}\right)^2+2\left(\frac{x}{l}\right)^3 \\ x-2l\left(\frac{x}{l}\right)^2+l\left(\frac{x}{l}\right)^3 \\ 3\left(\frac{x}{l}\right)^2-2\left(\frac{x}{l}\right)^3 \\ -l\left(\frac{x}{l}\right)^2+l\left(\frac{x}{l}\right)^3 \end{bmatrix} dx$$

$$F_3 = \int_0^l \left(\frac{3}{l^2}x^2 - \frac{1}{l^3}x^3\right)dx = \frac{ql}{2}$$

$$F_4 = \int_0^l \left(-\frac{x^2}{l} + \frac{x^3}{l^2}\right)dx = -\frac{ql^2}{12}$$

$$\frac{EI}{l^3}\left[\begin{array}{cc|cc} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 4l^2 \\ \hline -12 & -6l & 12 & -6l \\ 6l & 4l^2 & -6l & 4l^2 \end{array}\right]\begin{bmatrix} d_1 = 0 \\ d_2 = 0 \\ d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ \frac{ql}{2} \\ -\frac{ql^2}{12} \end{bmatrix}$$

$$\frac{EI}{l^3}\begin{bmatrix} 12 & -6l \\ -6l & 4l^2 \end{bmatrix}\begin{bmatrix} d_3 \\ d_4 \end{bmatrix} = \begin{bmatrix} \frac{ql}{2} \\ -\frac{ql^2}{12} \end{bmatrix}$$

$$\begin{bmatrix} d_3 \\ d_4 \end{bmatrix} = \frac{l^3}{EA}\frac{1}{12\times 4l^2 - 36l^2}\begin{bmatrix} 4l^2 & 6l \\ 6l & 12 \end{bmatrix}\begin{bmatrix} \frac{ql}{2} \\ -\frac{ql^2}{12} \end{bmatrix} = \begin{bmatrix} \frac{ql^4}{8EI} \\ \frac{ql^3}{6EI} \end{bmatrix}$$

$$w(x) = \vec{\phi}\cdot\vec{d}$$

Virtual displacement $\delta w = \vec{\phi}^T\delta\vec{d}$

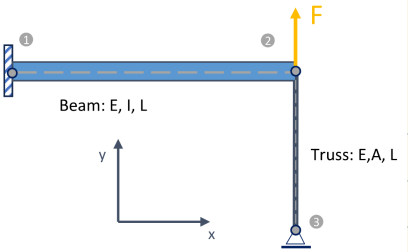
$$dF = q\,dx$$

$$d(\delta W) = dF\cdot\delta w = q\vec{\phi}^T\delta\vec{d}\,dx$$

$$\delta W = \int_0^l q\vec{\phi}^T\delta\vec{d}\,dx = \left(\int_0^l q\vec{\phi}^T dx\right)\delta\vec{d}$$

$$\delta W = F\cdot\delta d$$

$$\boxed{F = \int_0^l q\vec{\phi}^T dx}$$



Task 1: Find the displacement of node 2.
Task 2: Find the bending moment in node 1.
Task 3: Calculate the stress in the truss.

Element 1

$$K_e = \frac{EI}{l^3}\begin{bmatrix} 12 & 6l & -12 & 6l \\ 6l & 4l^2 & -6l & 2l^2 \\ -12 & -6l & 12 & -6l \\ 6l & 2l^2 & -6l & 4l^2 \end{bmatrix} \begin{matrix} 1Y \\ 1\theta \\ 2Y \\ 2\theta \end{matrix}$$

$$\frac{EI}{l^3}\begin{bmatrix} -12 & -6l \\ 6l & 4l^2 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ Y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} Y_2 \\ M_2 \end{bmatrix}$$

$$\begin{bmatrix} \frac{12EI+l^2EA}{l^3} & -\frac{6EI}{l^2} \\ -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}\begin{bmatrix} Y_2 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} Y_2 = F \\ M_2 = 0 \end{bmatrix}$$

$$\frac{12EI+l^2EA}{l^3}Y_2 - \frac{6EI}{l^2}\theta_2 = F \qquad -\frac{6EI}{l^2}Y_2 + \frac{4EI}{l}\theta_2 = 0$$

$$\frac{3EI+l^2EA}{l^3}Y_2 = F \qquad \theta_2 = \frac{3}{2l}Y_2$$

$$Y_2 = \frac{Fl^3}{3EI+l^2EA} \qquad \theta_2 = \frac{3Fl^2}{6EI+2l^2EA}$$

$$M_1 = \frac{EI}{l^3}\begin{bmatrix} 6l & 4l^2 & -6l & 2l^2 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ Y_2 \\ \theta_2 \end{bmatrix} = \frac{EI}{l^3}(-6lY_2 + 2l^2\theta_2) = \frac{3FI}{3I+l^2A}$$

$$Y_3 = \frac{EA}{l}\begin{bmatrix} 0 & 1 & 0 & -1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ Y_2 \\ \theta_2 \end{bmatrix} = -\frac{EA}{l}Y_2 = -\frac{Fl^2A}{3I+l^2A} \qquad N_3 = -Y_3 = \frac{FAl^2}{3I+Al^2}$$

Element 2 $S=1$ $C=0$

$$K_e = \frac{EA}{l}\begin{bmatrix} C^2 & SC & -C^2 & -SC \\ SC & S^2 & -SC & -S^2 \\ -C^2 & -SC & C^2 & SC \\ -SC & -S^2 & SC & S^2 \end{bmatrix} = \frac{EA}{l}\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix} \begin{matrix} 3X \\ 3Y \\ 2X \\ 2Y \end{matrix}$$

$$\frac{EA}{l}\begin{bmatrix} 0 & -1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 0 \\ X_2 \\ Y_2 \end{bmatrix} = Y_2$$

4. Beam Elements - Local vs Global Coordinate System

2D frame: axial + bending $\Rightarrow$ truss + beam element:

$$\begin{Bmatrix} d_1 \\ d_2 \\ d_3 \\ d_4 \\ d_5 \\ d_6 \end{Bmatrix} \Rightarrow \begin{Bmatrix} u_1 \\ v_1 \\ \theta_1 \\ u_2 \\ v_2 \\ \theta_2 \end{Bmatrix}$$

$$[k] = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{l^3} & \frac{6EI}{l^2} & 0 & -\frac{12EI}{l^3} & \frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{4EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{2EI}{l} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{l^3} & -\frac{6EI}{l^2} & 0 & \frac{12EI}{l^3} & -\frac{6EI}{l^2} \\ 0 & \frac{6EI}{l^2} & \frac{2EI}{l} & 0 & -\frac{6EI}{l^2} & \frac{4EI}{l} \end{bmatrix}$$

$$[k]_g = [T]^T [k][T]$$

$$[T] = \begin{bmatrix} c & -s & 0 & 0 & 0 & 0 \\ s & c & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & c & -s & 0 \\ 0 & 0 & 0 & s & c & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \qquad \begin{aligned} s &:= \sin\phi \\ c &:= \cos\phi \end{aligned}$$

7. Mass Matrix

Kinetic Energy: $T = \frac{1}{2}\rho A \int_0^L \dot{v}^2 dx$

Displacement Field: $v = \bar{\varphi}^T \cdot \vec{d}$

Velocity: $\dot{v} = \bar{\varphi}^T \cdot \dot{\vec{d}}$

$$T = \frac{1}{2}mv^2 = \frac{1}{2}\rho A \int_0^L \dot{\vec{d}}\,\bar{\varphi}\,\bar{\varphi}^T \dot{\vec{d}}\, dx = \frac{1}{2}\rho A \dot{\vec{d}}^T \int_0^L \bar{\varphi}\cdot\bar{\varphi}^T dx \cdot \dot{\vec{d}}$$

$$T = \frac{1}{2}\dot{\vec{d}}^T \cdot [m] \cdot \dot{\vec{d}}$$

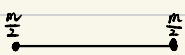
$$\boxed{[m] = \rho A \int_0^L \bar{\varphi}\cdot\bar{\varphi}^T dx}$$

Truss Element Consistent Mass Matrix

$\vec{d} = \begin{bmatrix} u_{1x} \\ u_{2x} \end{bmatrix}$ $\quad \bar{\varphi} = \begin{bmatrix} \phi_1(x) \\ \phi_2(x) \end{bmatrix} = \begin{bmatrix} 1 - \frac{x}{l} \\ \frac{x}{l} \end{bmatrix}$

$$[m]_t = \frac{\rho A l}{6}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Truss Element Isotropic Lumped Mass Matrix



$$[m] = \frac{\rho A l}{2}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \qquad [m]_t = \frac{\rho A l}{2}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Truss Element Rotated to Global coordinates

$$[m]_t = \rho A l \begin{bmatrix} \frac{1}{3}c^2 & \frac{1}{3}sc & \frac{1}{6}c^2 & \frac{1}{6}sc \\ \frac{1}{3}sc & \frac{1}{3}s^2 & \frac{1}{6}sc & \frac{1}{6}s^2 \\ \frac{1}{6}c^2 & \frac{1}{6}sc & \frac{1}{3}c^2 & \frac{1}{3}sc \\ \frac{1}{6}sc & \frac{1}{6}s^2 & \frac{1}{3}sc & \frac{1}{3}s^2 \end{bmatrix}$$

Beam Element Consistent Mass Matrix

$$[m] = \frac{\rho A l}{420}\begin{bmatrix} 156 & 22l & 54 & -13l \\ 22l & 4l^2 & 13l & -3l^2 \\ 54 & 13l & 156 & -22l \\ -13l & -3l^2 & -22l & 4l^2 \end{bmatrix}$$

Structural Dynamics

Structural dynamics: $[m][\ddot{d}] + [c][\dot{d}] + [k][d] = f(t)$

Modal Analysis: eigenfrequency, eigenmodes

Frequency response Analysis: system response due to harmonic excitation

Direct time integration: arbitrary excitation

Important in: safety (failure due to undamped response)

comfort (vibrations)

vehicle control

$[m][\ddot{d}]+[k][d]=0$ (no damping, no external force)

Euler's identity: $e^{iwt}=\cos(wt)+i\sin(wt)$

t: time angular frequency: w amplitude vector: $A\cdot\vec{\phi}$ $\vec{\phi}$: mode shape vector

Assume: $d=A\cdot\vec{\phi}e^{iwt}$

$\dot{d}=A\cdot\vec{\phi}iwe^{iwt}$

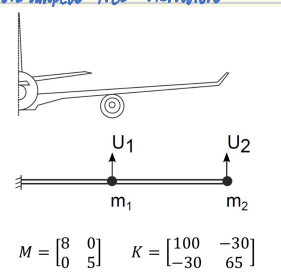
$\ddot{d}=-A\cdot\vec{\phi}w^2e^{iwt}$

$-[m](A\cdot\vec{\phi}w^2e^{iwt})+[k]A\cdot\vec{\phi}e^{iwt}=0$

$([k]-w^2[m])A\cdot\vec{\phi}e^{iwt}=0$

$\det([k]-w^2[m])=0$ $w^2=\lambda$

Undamped Free Vibration



$M=\begin{bmatrix}8 & 0\\ 0 & 5\end{bmatrix}$ $K=\begin{bmatrix}100 & -30\\ -30 & 65\end{bmatrix}$

Step 1: Dynamic Modelling Step 2: Assume harmonic solution

$[m][\ddot{d}]+[K][d]=0$ $d(t)=A\phi e^{iwt}$

Step 3: Solution of eigenvalues

$\det([K]-\lambda[m])=0$

$\begin{vmatrix}100-8\lambda & -30\\ -30 & 65-5\lambda\end{vmatrix}=0$

$\lambda_1=8$ $\lambda_2=175$ (Lower first)

$w_1=\sqrt{\lambda_1}$ $w_2=\sqrt{175}$ (Hz)

Eigenfrequency $f_1=\frac{w_1}{2\pi}=0.45$ $f_2=\frac{w_2}{2\pi}=0.666$ (rad/s)

Step 4: Eigenvector/shape vector ϕ_i

$([K]-\lambda_i[m])\phi_i=0$

Mode 1: $\lambda_1=8$

$\begin{bmatrix}36 & -30\\ -20 & 25\end{bmatrix}\begin{bmatrix}\phi_{1,1}\\ \phi_{1,2}\end{bmatrix}=0$

$\phi_{1,1}=1$ $\phi_{1,2}=\frac{6}{5}$ $\vec{\phi}_1=\begin{bmatrix}1\\ \frac{6}{5}\end{bmatrix}$

Mode 2: $\lambda_2=175$

$\begin{bmatrix}-40 & -30\\ -30 & -22.5\end{bmatrix}\begin{bmatrix}\phi_{2,1}\\ \phi_{2,2}\end{bmatrix}=0$

$\phi_{2,1}=1$ $\phi_{2,2}=-\frac{4}{3}$ $\vec{\phi}_2=\begin{bmatrix}1\\ -\frac{4}{3}\end{bmatrix}$

mass normalization: $[\theta_i]^T[m][\theta_i]=1$

$[a\phi_{i,1}\ \ a\phi_{i,2}]\begin{bmatrix}8&0\\0&5\end{bmatrix}\begin{bmatrix}a\phi_{i,1}\\a\phi_{i,2}\end{bmatrix}=1$

$8a_i^2\phi_{i,1}^2+5a_i^2\phi_{i,2}^2=1$

$a_1=\sqrt{\frac{1}{8\phi_{1,1}^2+5\phi_{1,2}^2}}=\frac{\sqrt{95}}{38}$ $\quad a_2=\sqrt{\frac{1}{8\phi_{2,2}^2+5\phi_{2,2}^2}}=\frac{3\sqrt{57}}{76}$

$\phi_{1,n}=\frac{\sqrt{95}}{38}\begin{bmatrix}1\\\frac{8}{5}\end{bmatrix}$ $\quad \phi_{2,n}=\frac{3\sqrt{38}}{76}\begin{bmatrix}1\\-\frac{4}{3}\end{bmatrix}$

$[\theta]=[\phi_{1,n}\ \phi_{2,n}]=\begin{bmatrix}\frac{\sqrt{95}}{38}&\frac{3\sqrt{57}}{76}\\\frac{8\sqrt{95}}{190}&-\frac{\sqrt{38}}{19}\end{bmatrix}$

$M_i=\phi_i^T[M]\phi_i$

$\phi_{i,n}=\frac{\phi_i}{\sqrt{\phi_i^T[M]\phi_i}}$

Transformation to Generalized Coordinates

Structure dynamics: $[m][\ddot{d}]+[K][d]=F(t)$

Approach: $[d]=[\phi][q]=\sum_{i=1}^{n}[\phi_i]q_i$

$[\phi]^T[m][\phi]\ddot{q}+[\phi]^T[K][\phi]q=[\phi]^T[F]$

generalized mass matrix → $\vec{I}$; generalized stiffness matrix → $\vec{\Lambda}$; generalized force vector → $\vec{Q}$

$\begin{bmatrix}1&0\\0&1\end{bmatrix}[\ddot{q}]+\begin{bmatrix}w_1^2&0\\0&w_2^2\end{bmatrix}[q]=[Q]$

7. Summary Modal Analysis and Modal Transformation

$[M]\{\ddot{x}\}+[K]\{x\}=0$

n number of degrees of freedom (DoF) $\quad \{x\}$ physical coordinates vector

$[M]$ mass matrix $n\ x\ n$ symmetric $[M]=[M]^T$ positive definite $\{y\}^T[M]\{y\}>0$

In mechanical systems the mass matrix (from FEM) is always symmetric and positive definite.

$[K]$ stiffness matrix $n\ x\ n$ symmetric $[K]=[K]^T$ positive definite $\{y\}^T[K]\{y\}>0$

In linear elastic mechanical systems, the stiffness matrix (from FEM) is always symmetric and positive definite for sufficient support boundary conditions.

$(K-\lambda M)\{\Phi\}=0$ general eigenvalue problem [M] ≠ [I] (identity matrix, unit matrix)
For this problem: all eigenvalues real and positive
n DoFs -> n, real and positive eigenfrequencies $\quad \omega_i=\sqrt{\lambda_i}$

$(A-\lambda I)\{\Phi\}=0$ (for comparison: special eigenvalue problem for matrix A and unit matrix)

$[\Phi]=[\{\Phi_1\},\{\Phi_j\},\ldots,\{\Phi_n\}]$ modal matrix with n different linear independent eigenvectors

7. Summary Modal Analysis and Modal Transformation

$\{\Phi_i\}^T[M]\{\Phi_j\}=0\ \{\Phi_i\}\neq\{\Phi_j\}$ separate eigenvectors are orthogonal to M

physical interpretation: inertia forces from one eigenmode do not contribute to another eigenmode

$\{\Phi_i\}^T[K]\{\Phi_j\}=0\ \{\Phi_i\}\neq\{\Phi_j\}$ separate eigenvectors are orthogonal to K

therefore:

$[M_G]=[\Phi]^T[M][\Phi]$ $\quad [K_G]=[\Phi]^T[K][\Phi]$ are diagonal matrices

$\{x\}=[\Phi]\{q\}$ modal coordinates vector

$[\Phi_n]=\frac{1}{\sqrt{[\Phi]^T[M][\Phi]}}[\Phi]$ norming of modal matrix with respect to mass matrix

$[M_{G,n}]=[\Phi_n]^T[M][\Phi_n]=[I]$ normed mass matrix is identity matrix (quadratic form)

$[K_{G,n}]=[\Phi_n]^T[K][\Phi_n]=[\Lambda]$ normed stiffness matrix with eigenvalues on diagonal (quadratic form)